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Goldenberg et al.

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(54) **LARGE-APERTURE COMPACT SCANNING TELE CAMERAS**

(71) Applicant: **Corephotonics Ltd.**, Tel Aviv (IL)

(72) Inventors: **Ephraim Goldenberg**, Tel Aviv (IL); **Yiftah Kowal**, Tel Aviv (IL); **Gal Shabtay**, Tel Aviv (IL); **Itamar Boral**, Tel Aviv (IL); **Ziv Shemesh**, Tel Aviv (IL)

(73) Assignee: **Corephotonics Ltd.**, Tel Aviv (IL)

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(21) Appl. No.: **19/204,545**

(22) Filed: **May 11, 2025**

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(60) Provisional application No. 63/380,786, filed on Oct. 25, 2022, provisional application No. 63/297,256, filed on Jan. 7, 2022, provisional application No. 63/289,323, filed on Dec. 14, 2021.

(51) **Int. Cl.**
G02B 13/00 (2006.01)
G02B 13/02 (2006.01)
G02B 26/10 (2006.01)
G03B 13/36 (2021.01)
G03B 17/17 (2021.01)
H04N 23/54 (2023.01)

(52) **U.S. Cl.**
CPC **G02B 13/0065** (2013.01); **G02B 13/02** (2013.01); **G02B 26/108** (2013.01); **G03B 13/36** (2013.01); **G03B 17/17** (2013.01); **H04N 23/54** (2023.01)

(58) **Field of Classification Search**
CPC .. **G02B 13/0065**; **G02B 13/02**; **G02B 26/108**; **H04N 23/54**; **G03B 13/36**; **G03B 17/17**
USPC **359/745**
See application file for complete search history.

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359/220.1

* cited by examiner

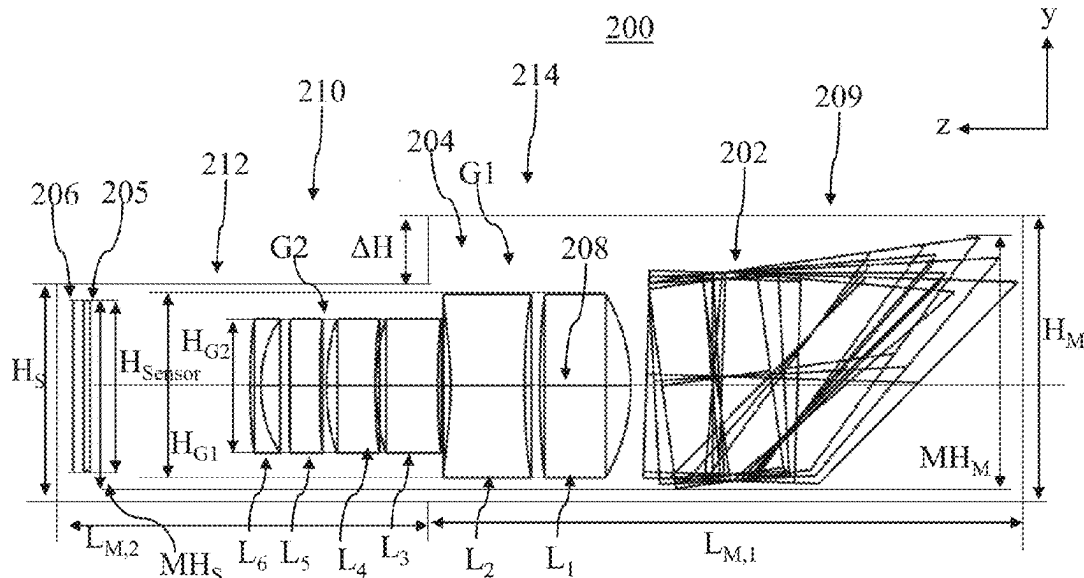
Primary Examiner — William Choi

(74) *Attorney, Agent, or Firm* — Nathan & Associates;
Menachem Nathan

(57) ABSTRACT

Scanning Tele cameras (STCs) based on two optical path folding element (OPFE) field-of-view scanning and mobile devices including such STCs. A STC may comprise a first OPFE (O-OPFE) for folding a first optical path OP1 to a second optical path OP2, an O-OPFE actuator, a second OPFE (I-OPFE) for folding OP2 to a third optical path OP3, an I-OPFE actuator, a lens, a lens actuator and an image sensor, wherein the STC has a STC native field-of-view ($n\text{-FOV}_T$), wherein the O-OPFE actuator is configured to rotate the O-OPFE around a first axis and the I-OPFE actuator rotates the I-OPFE around a second axis for scanning a scene with the $n\text{-FOV}_T$, wherein the lens actuator is configured to move the lens for focusing along a third axis, and wherein the first axis is perpendicular to the second axis and parallel to the third axis.

20 Claims, 21 Drawing Sheets



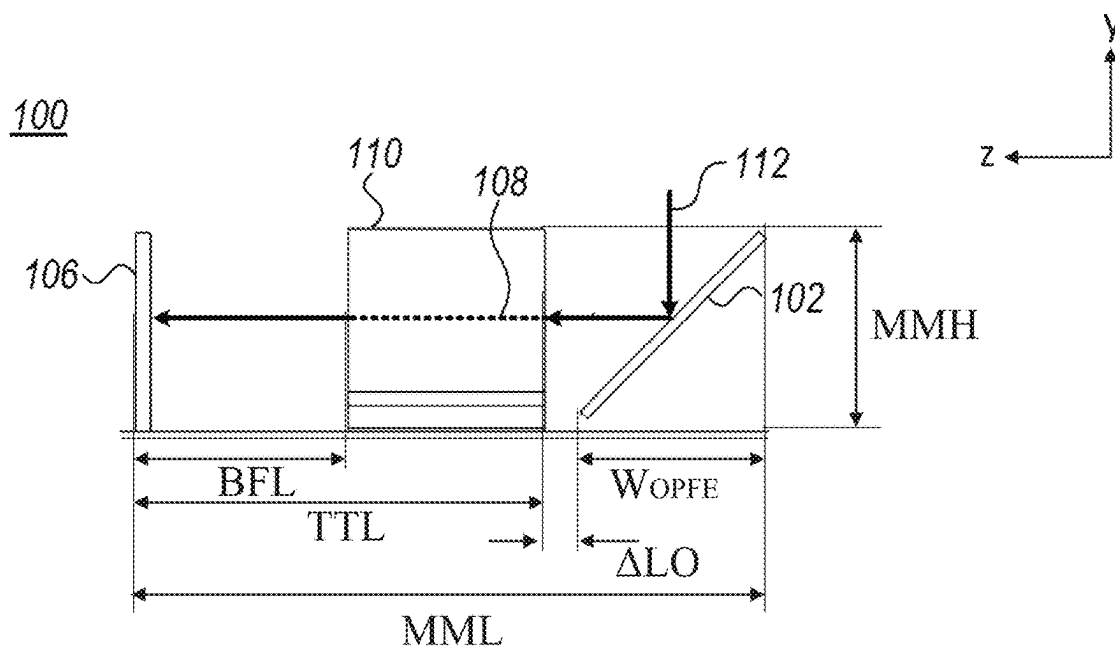


FIG. 1A KNOWN ART

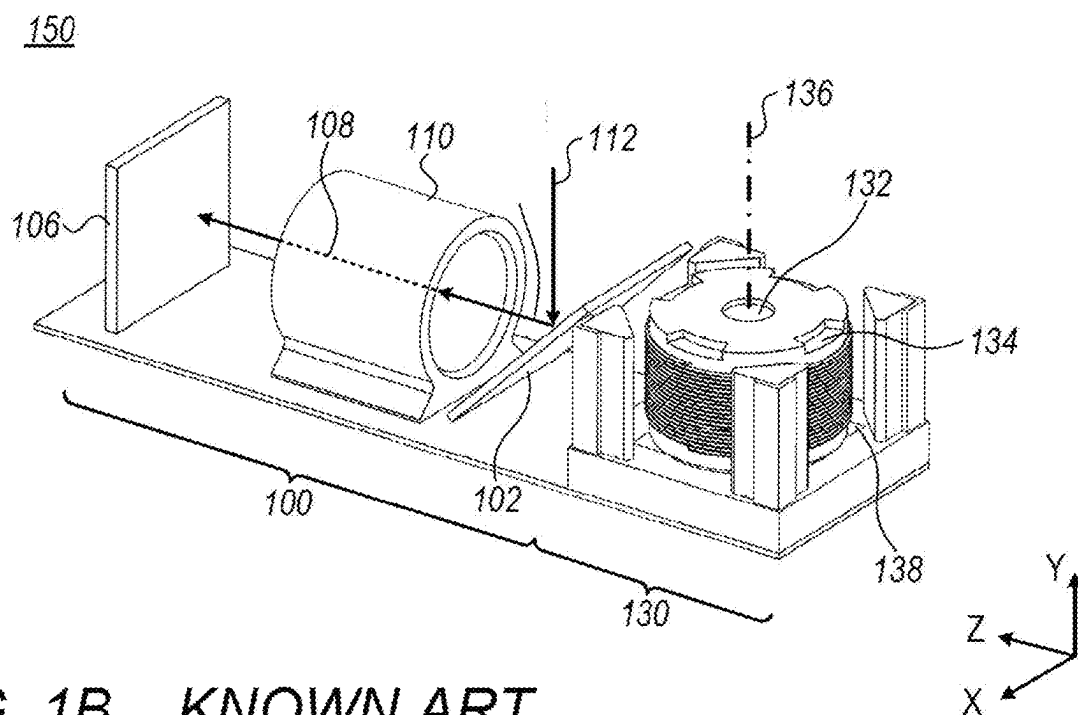
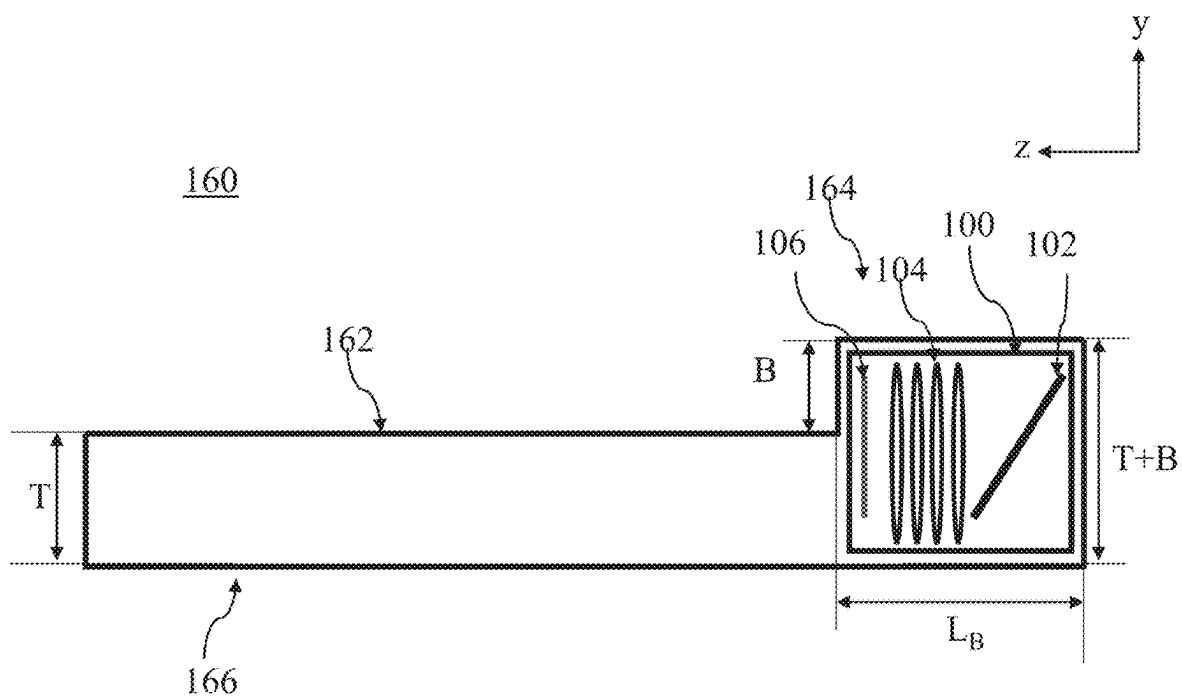


FIG. 1B KNOWN ART



KNOWN ART

FIG. 1C

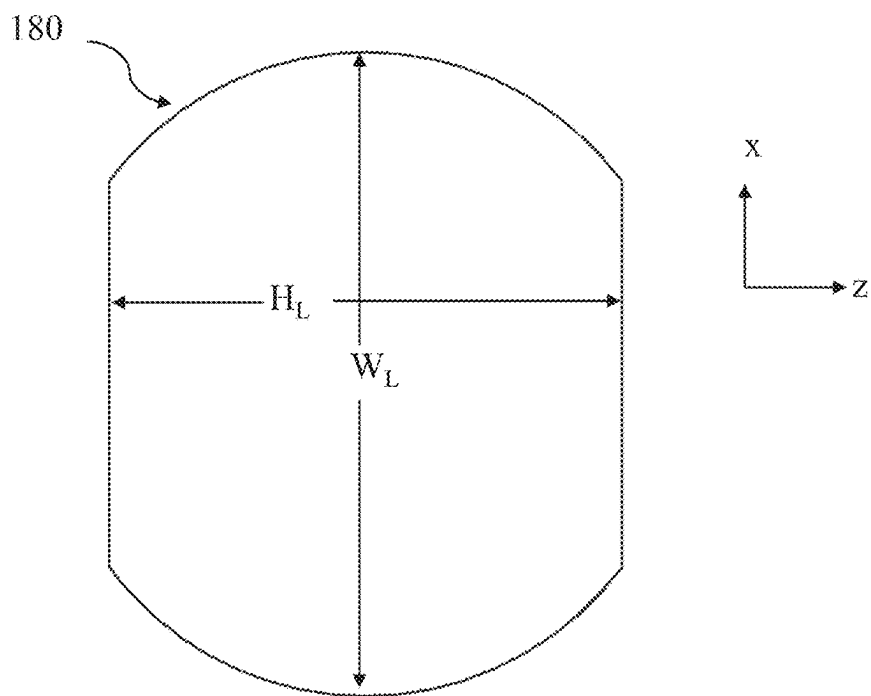


FIG. 1D

KNOWN ART

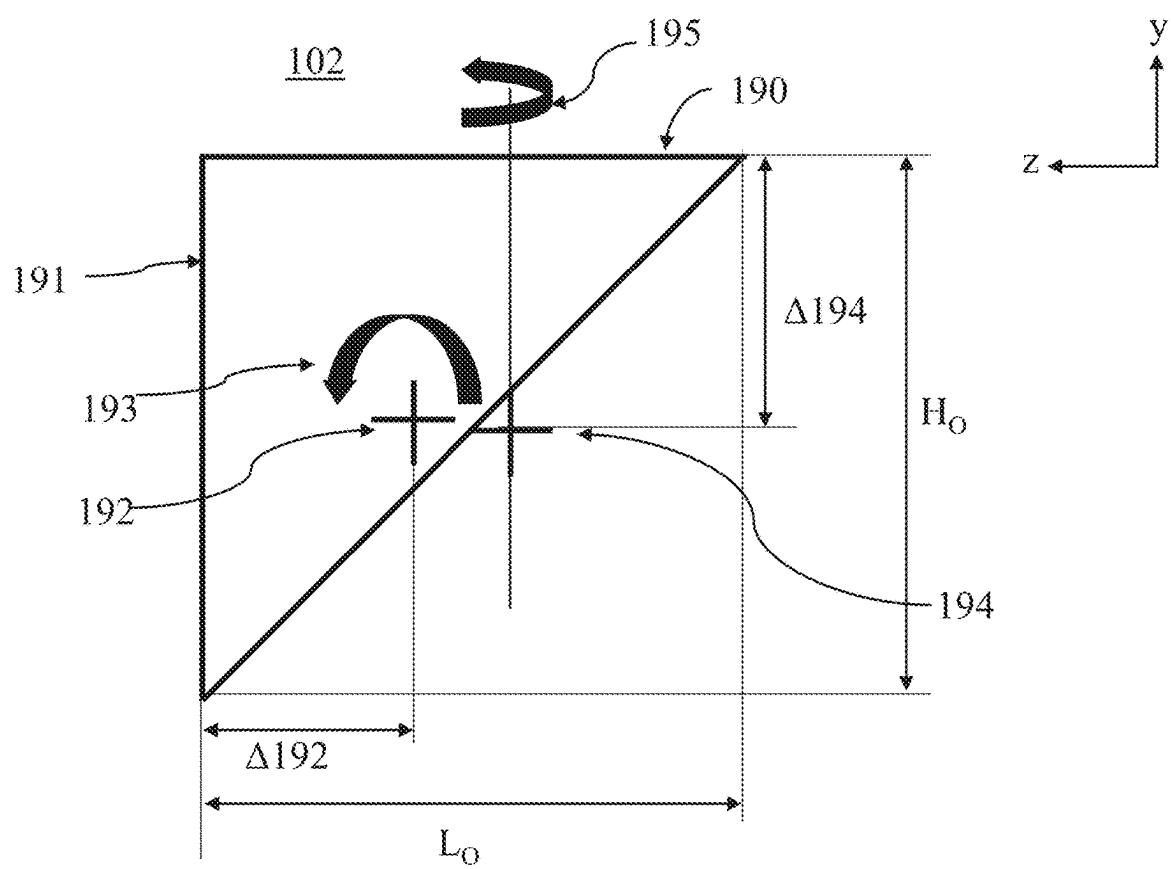


FIG. 1E

KNOWN ART

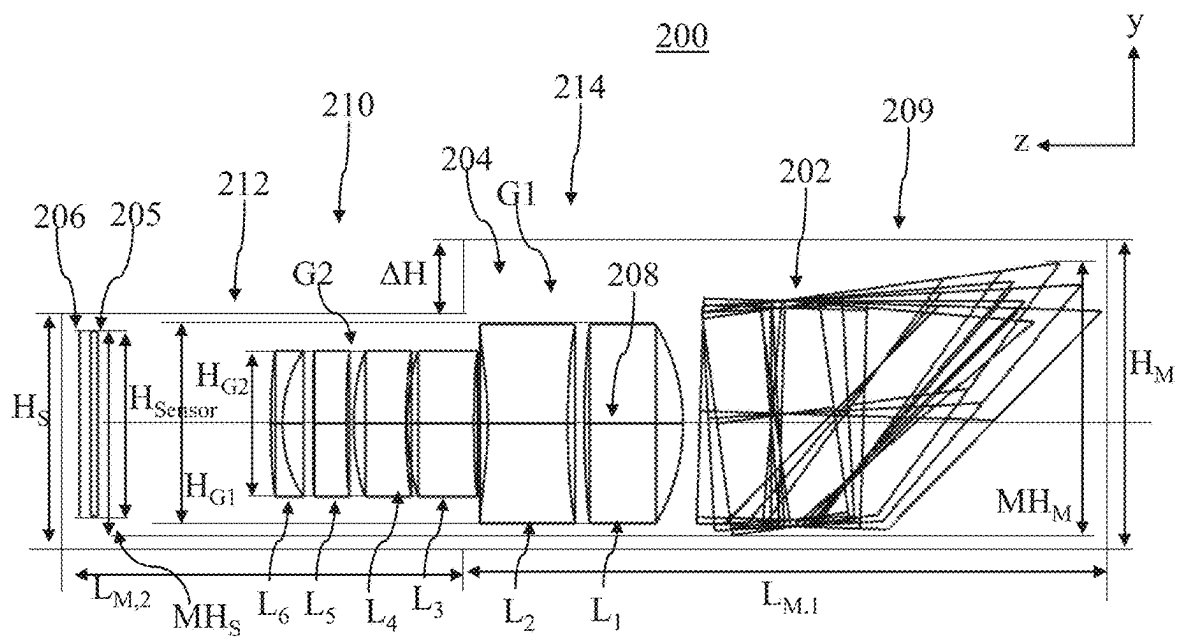


FIG. 2A

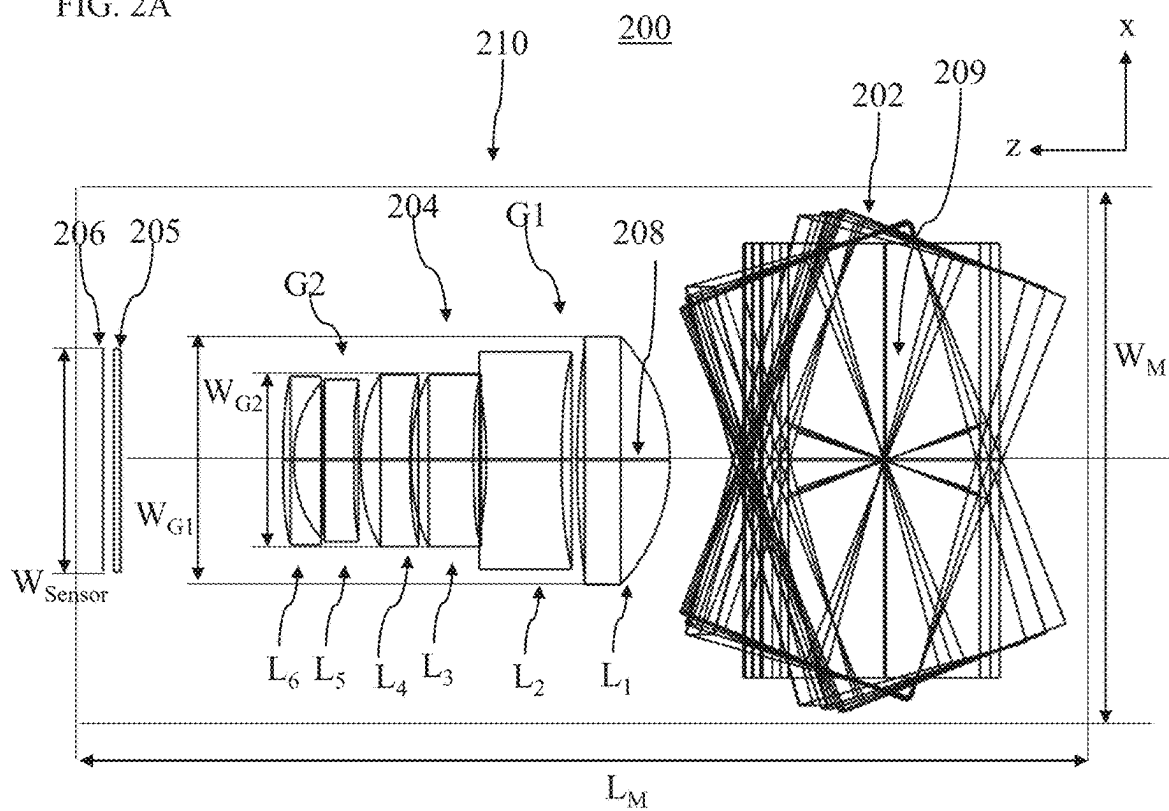


FIG. 2B

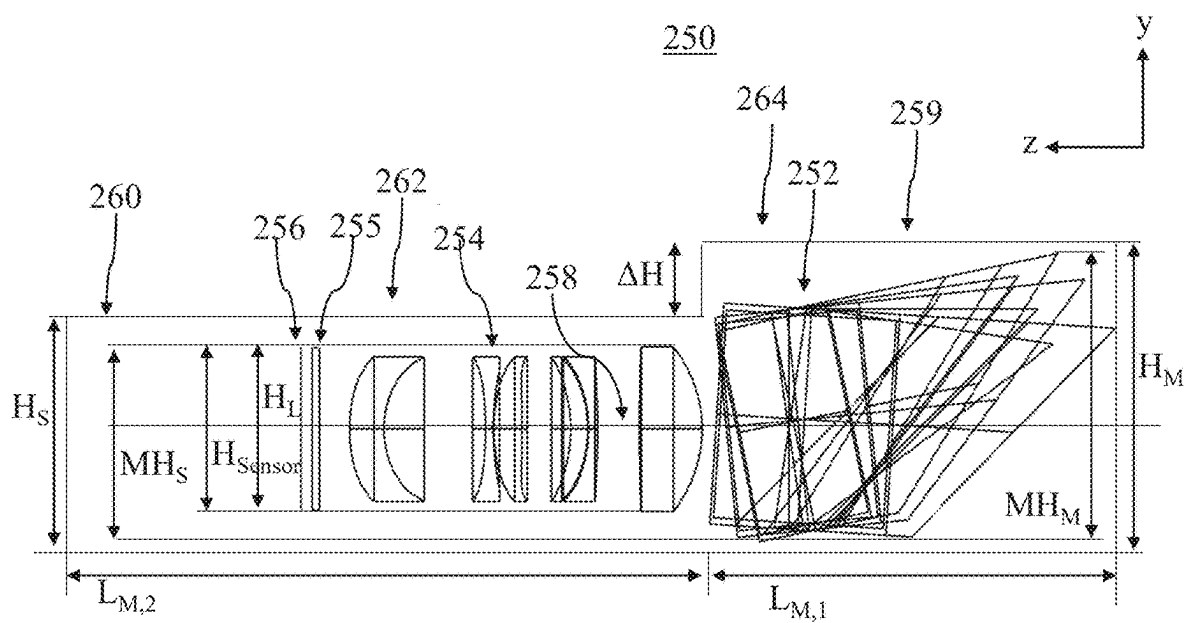


FIG. 2C

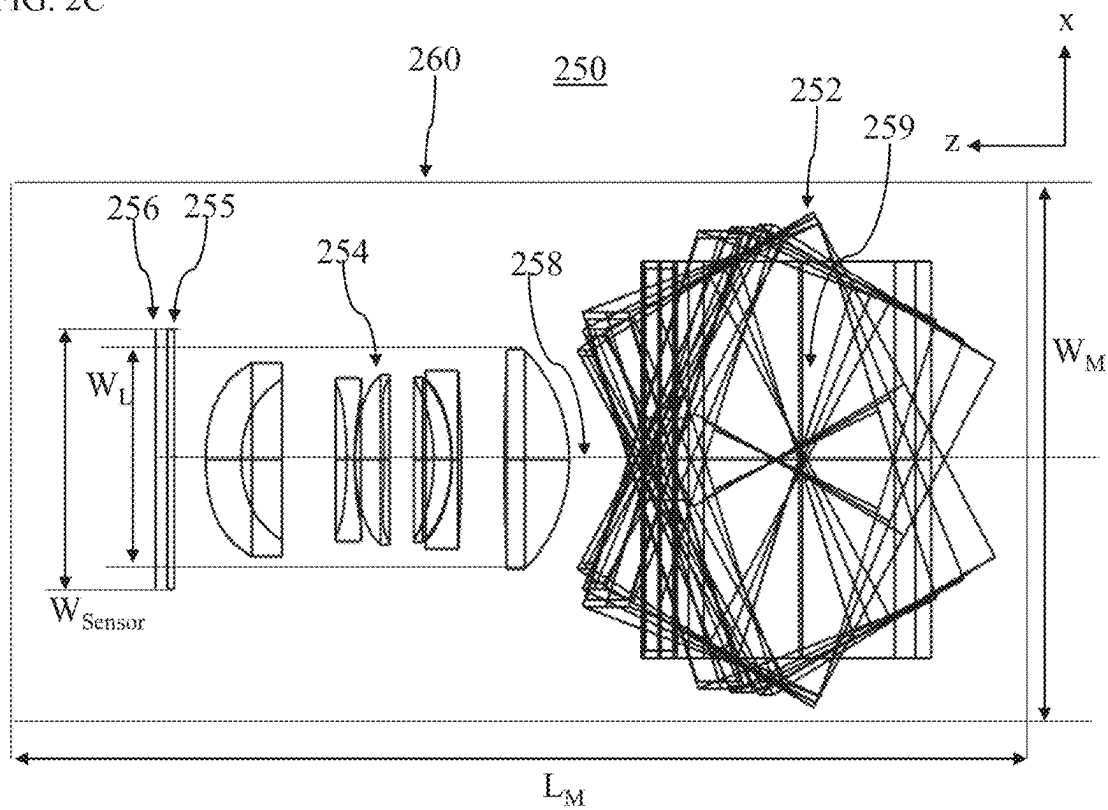
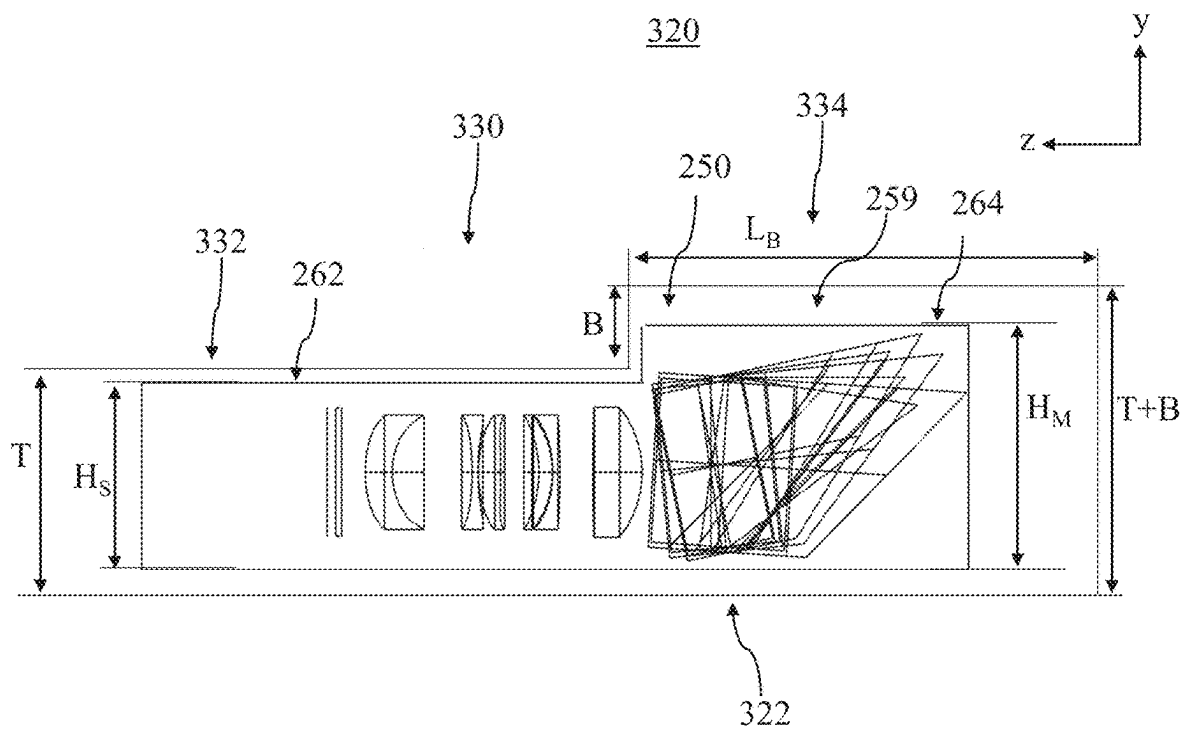
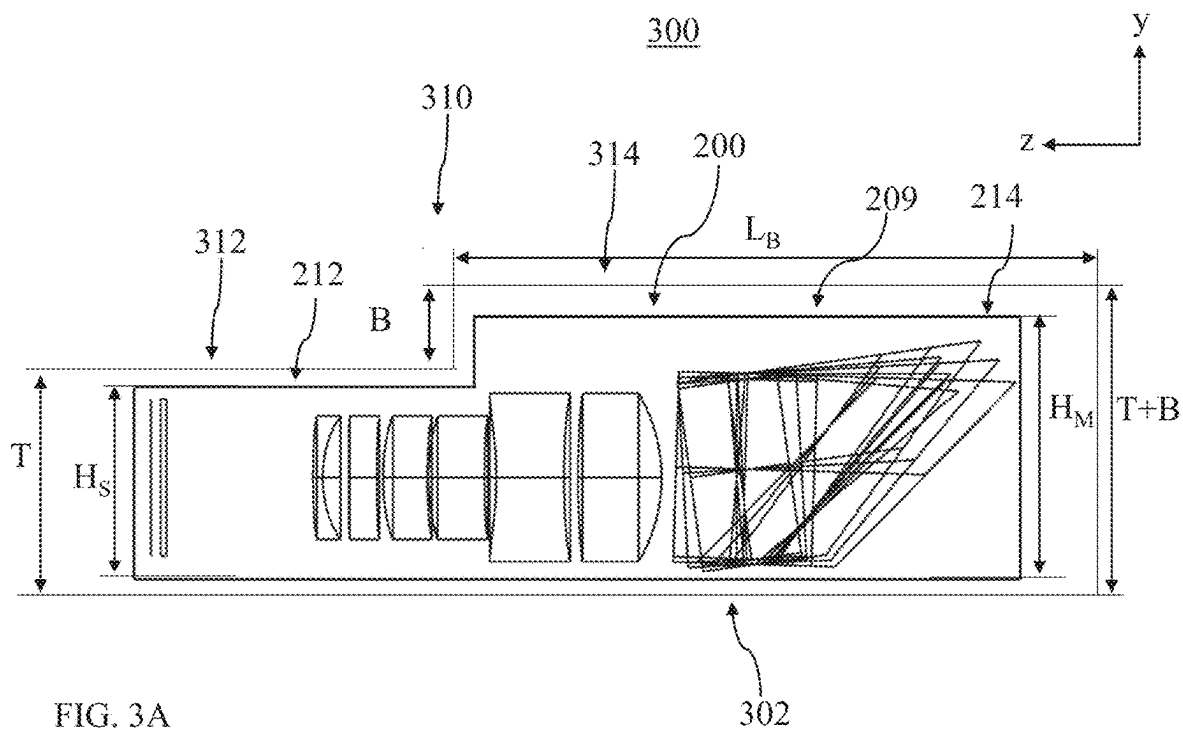


FIG. 2D



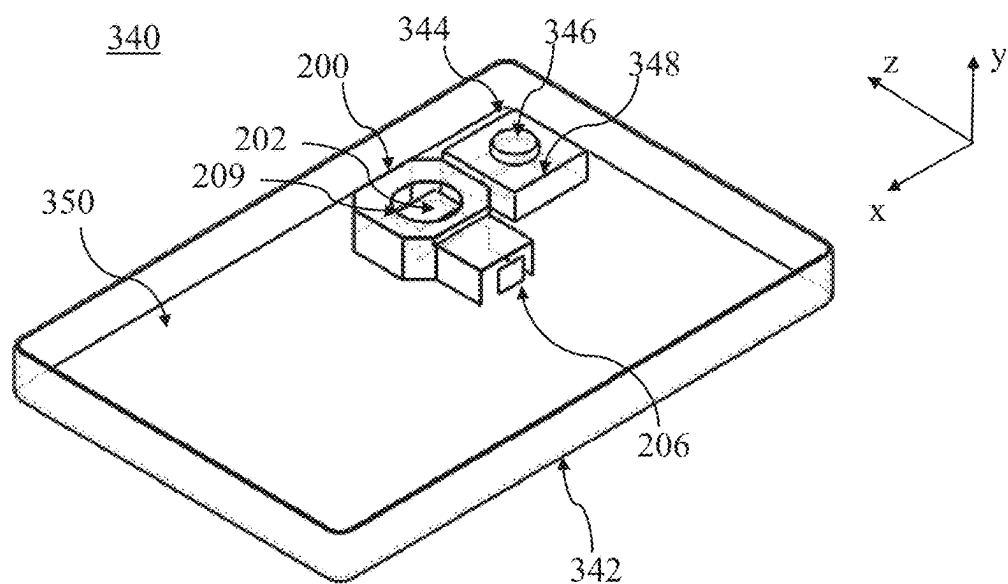


FIG. 3C

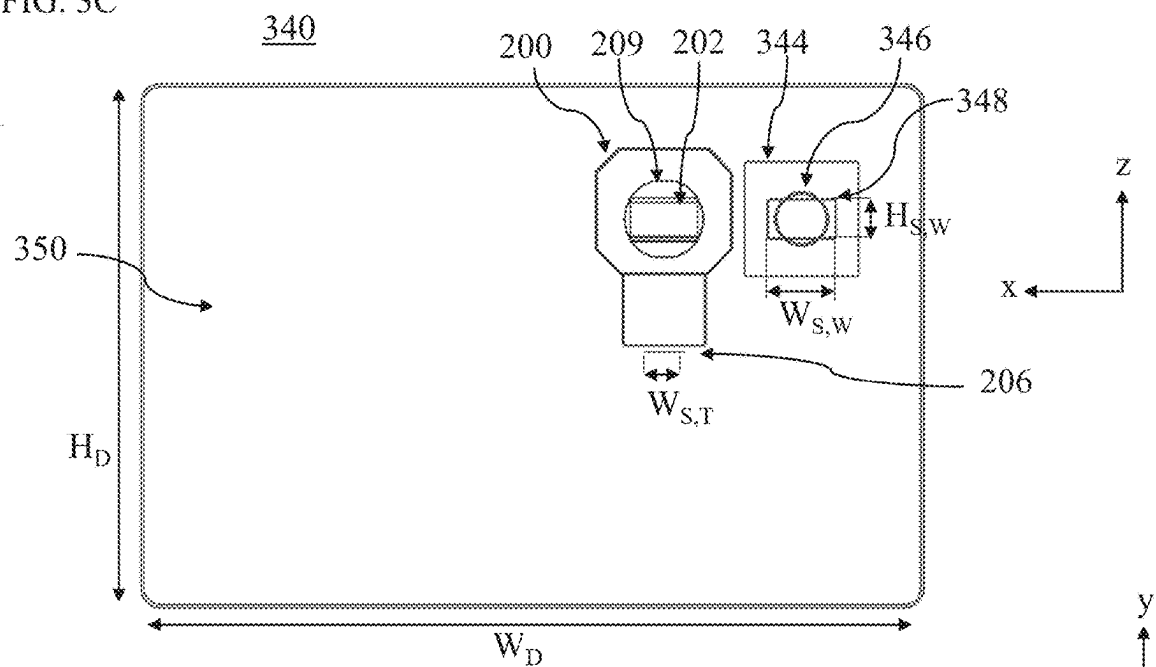


FIG. 3D

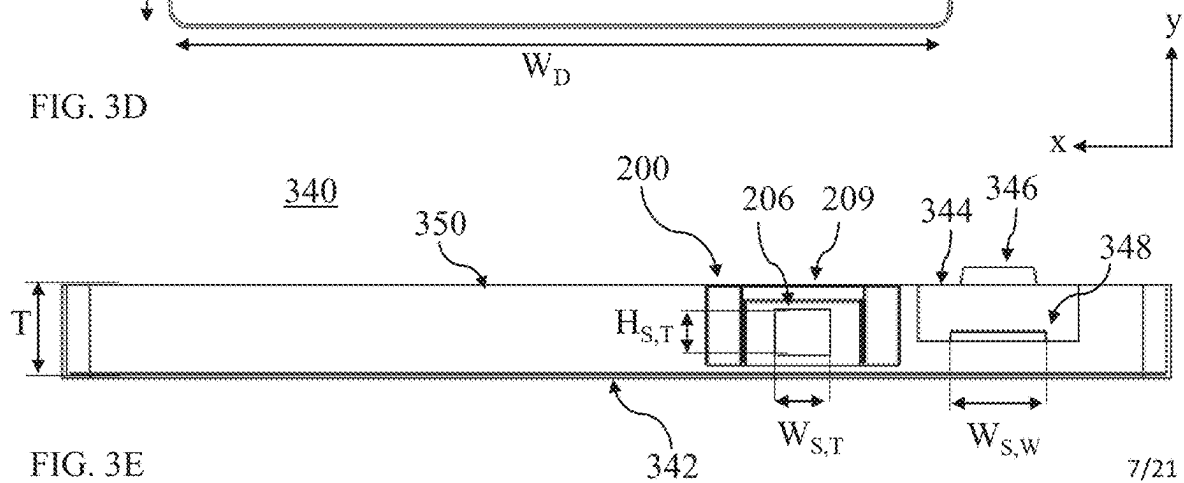


FIG. 3E

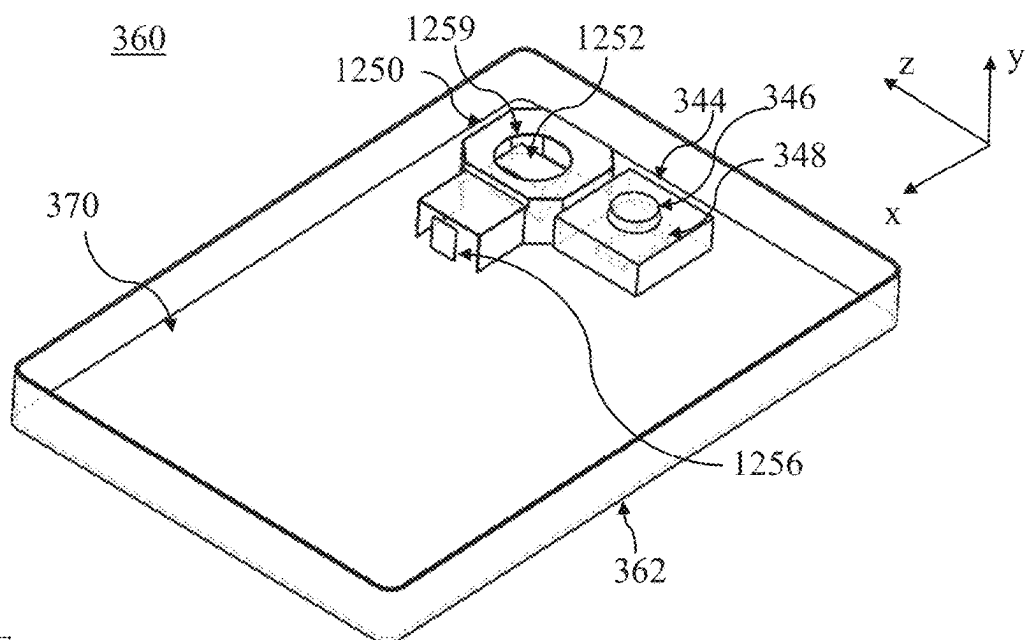


FIG. 3F

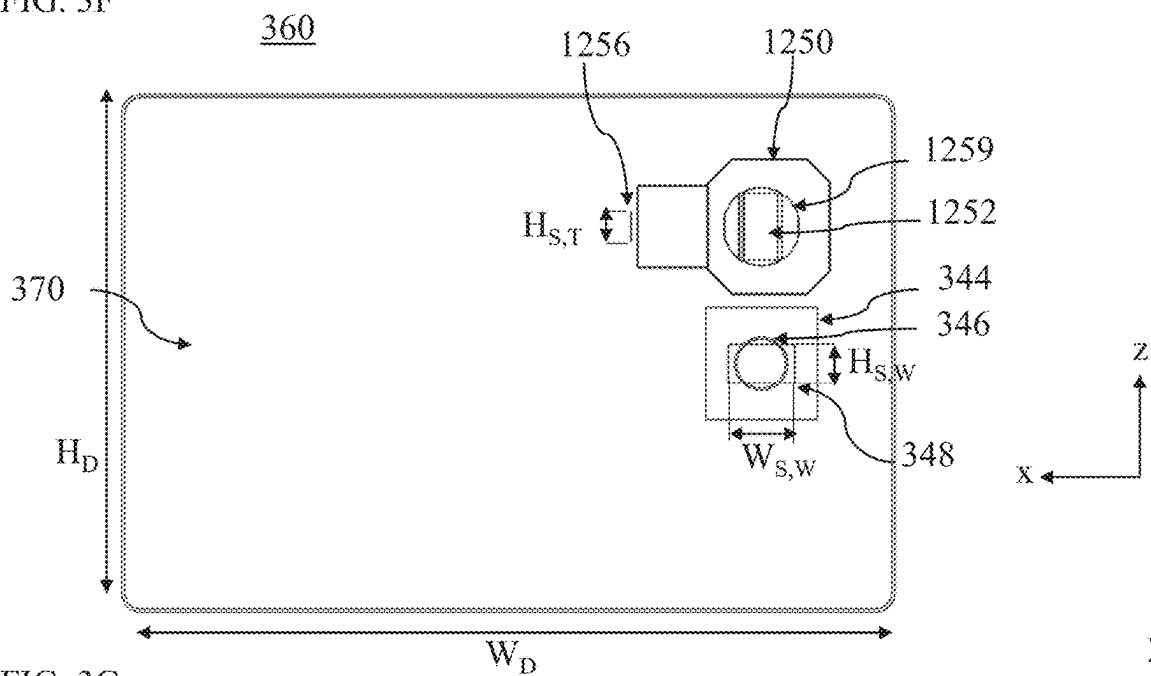


FIG. 3G

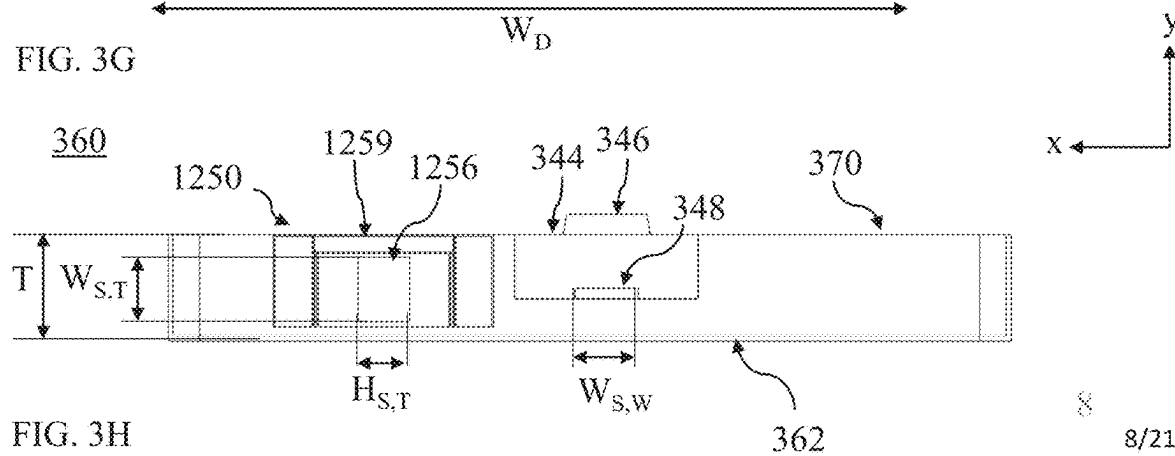
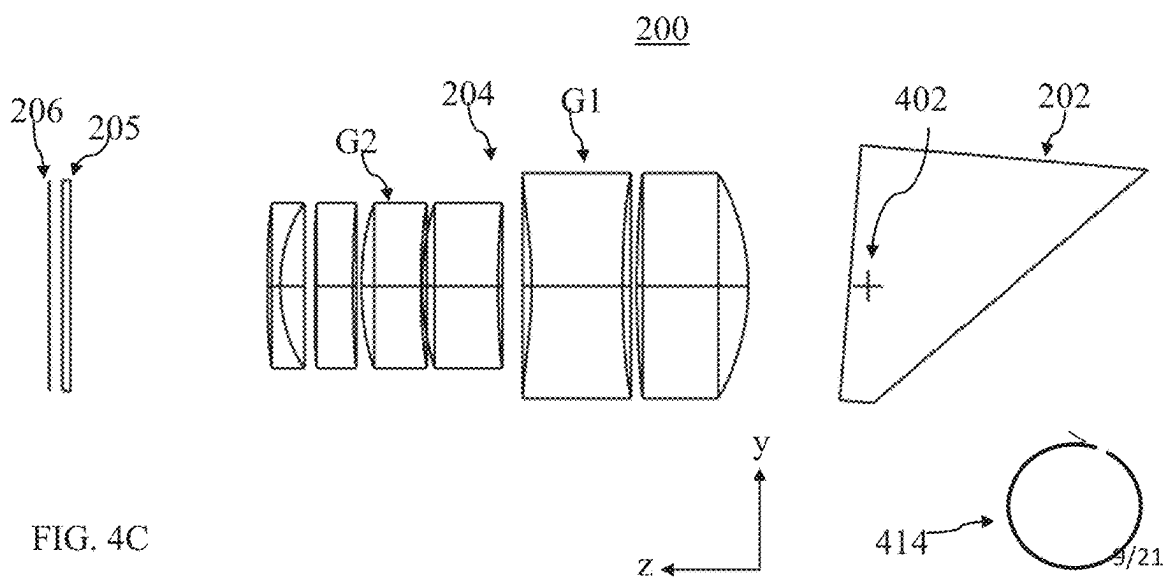
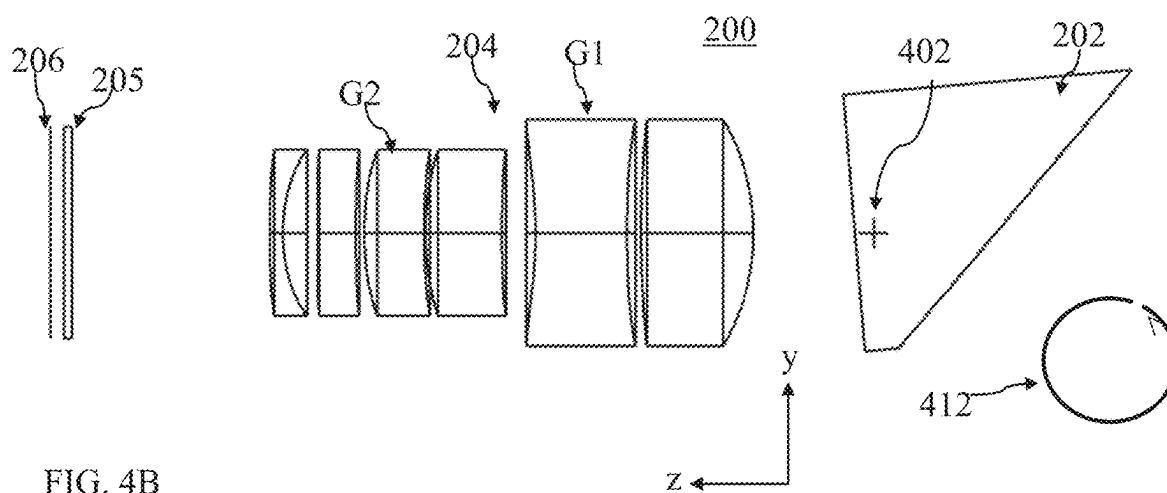
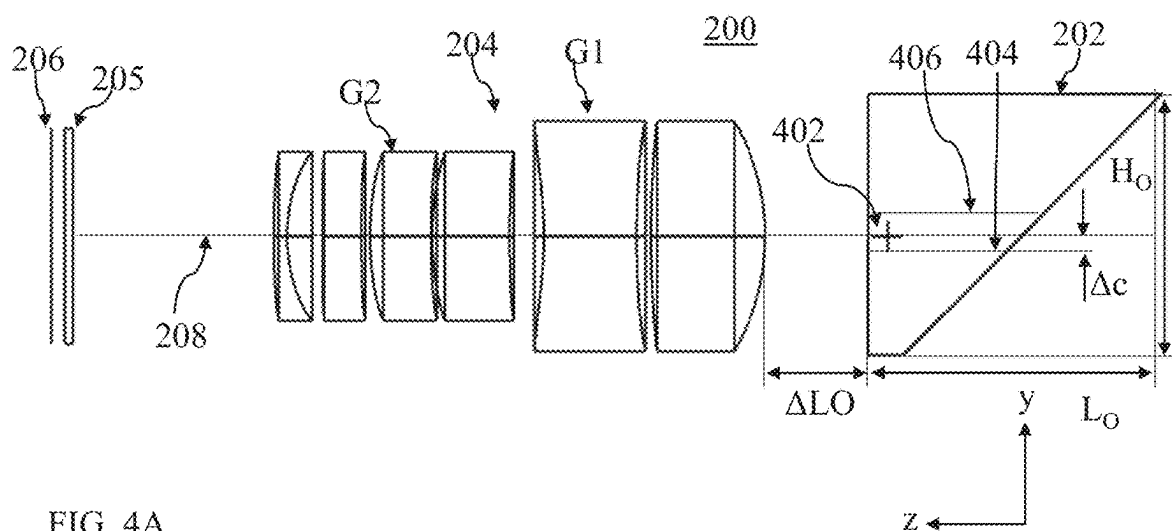


FIG. 3H



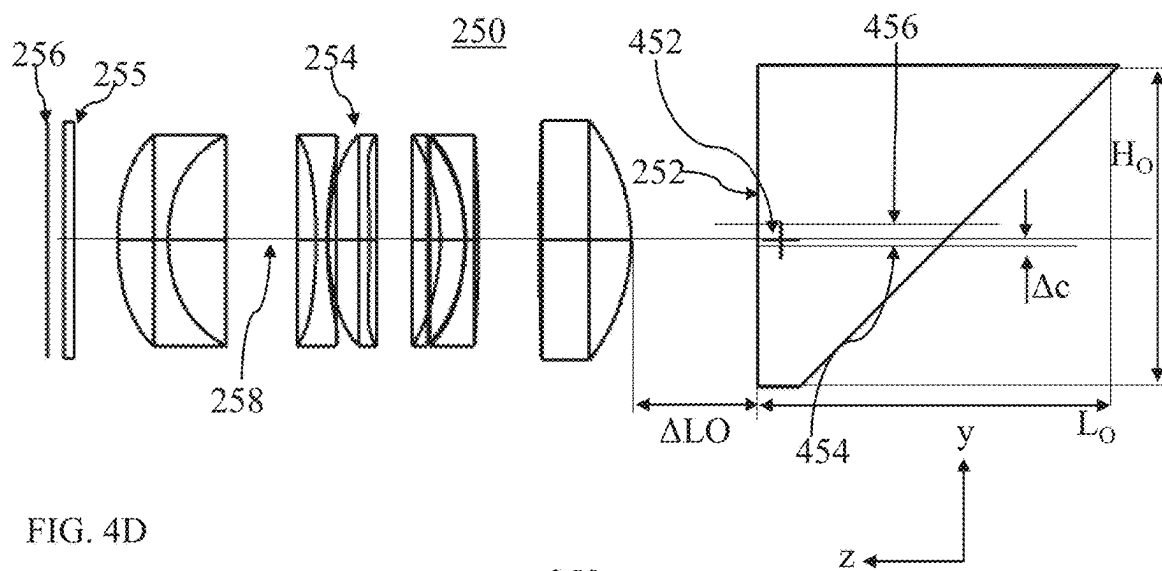


FIG. 4D

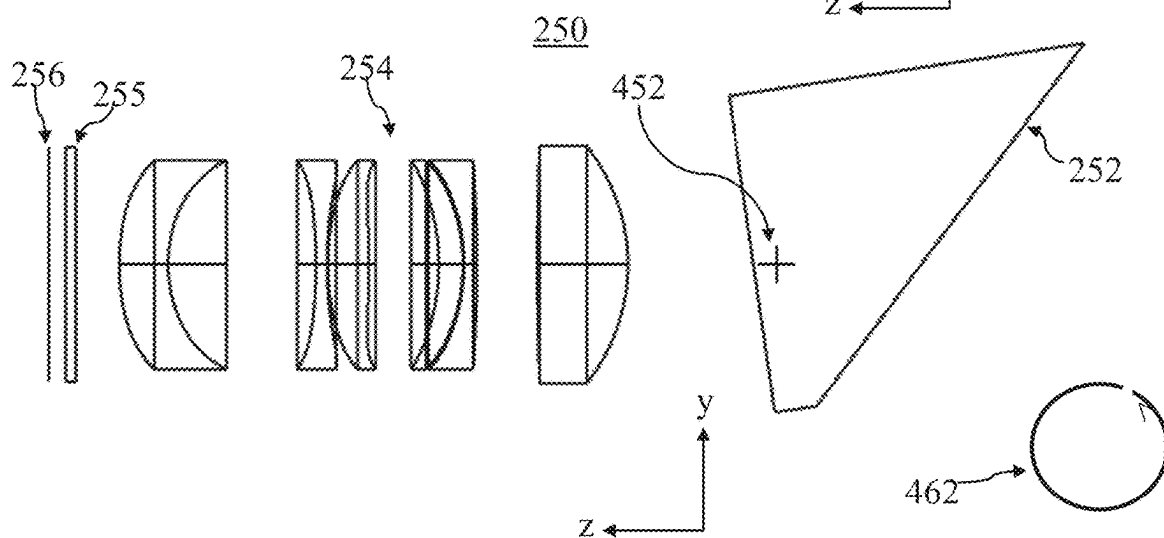


FIG. 4E

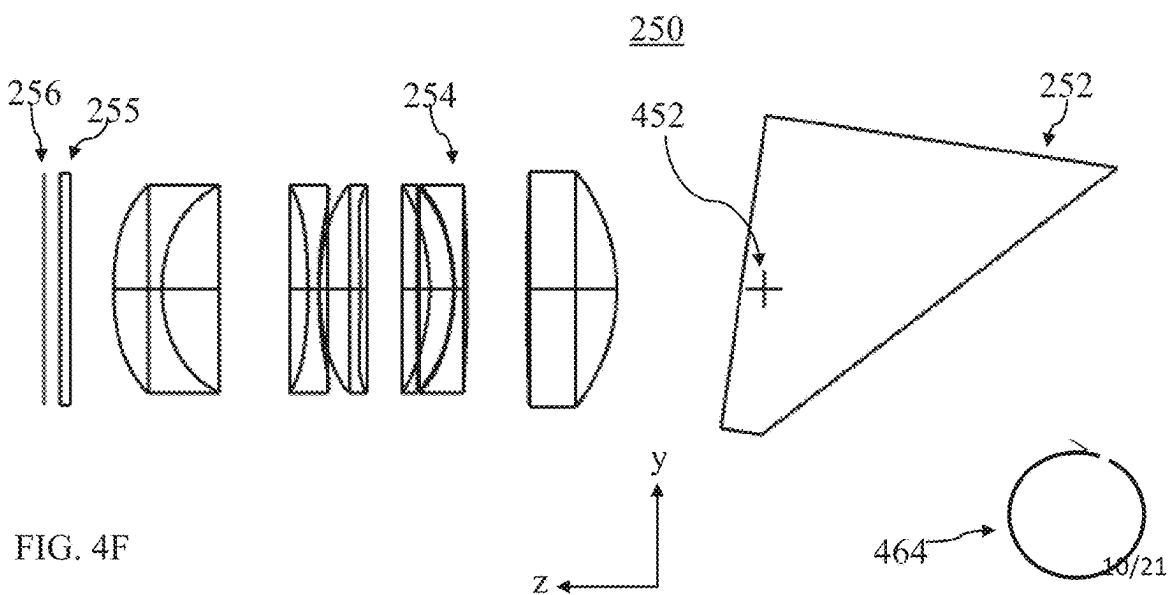


FIG. 4F

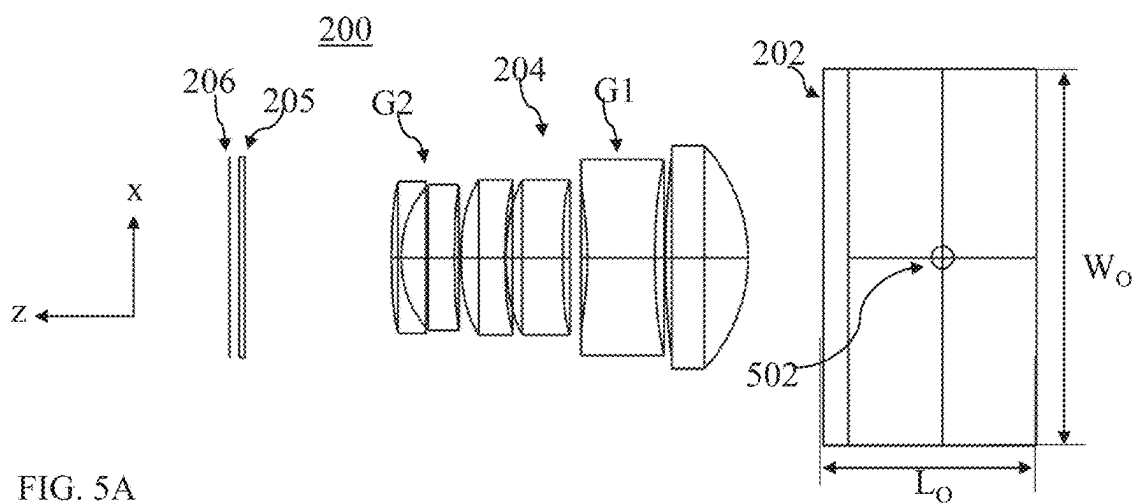


FIG. 5A

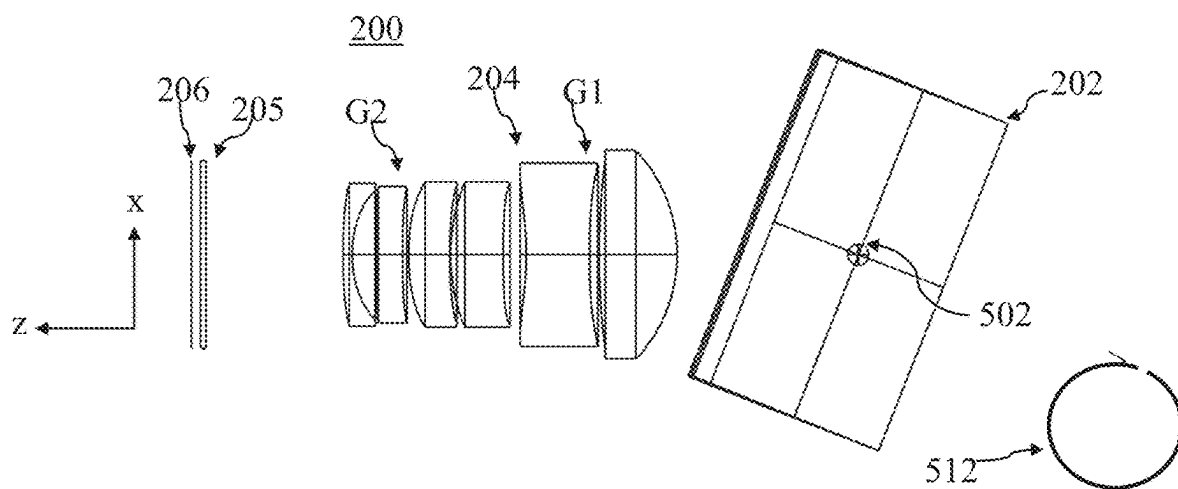


FIG. 5B

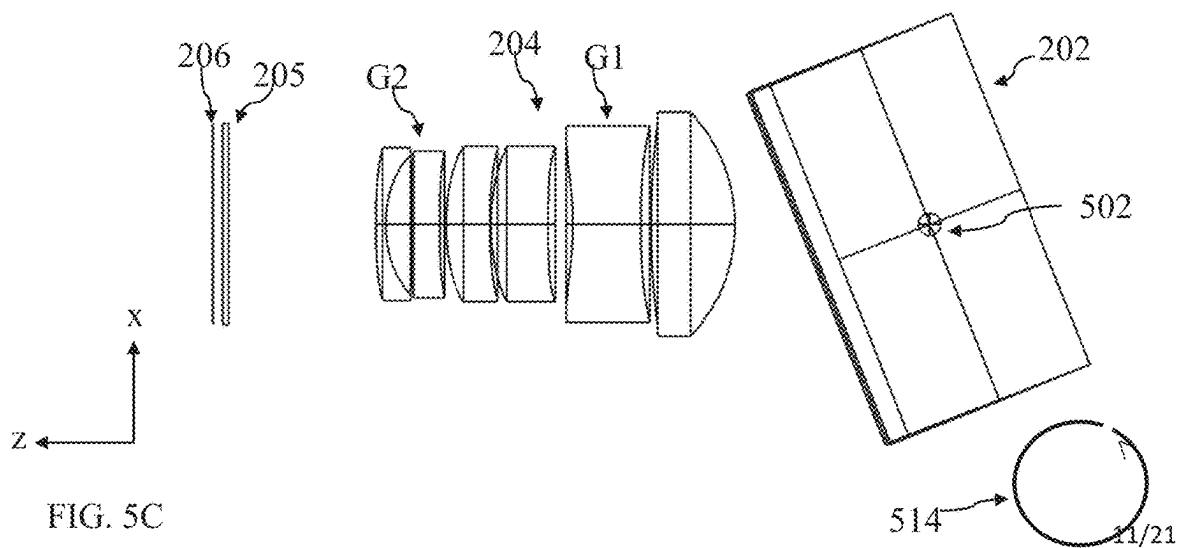
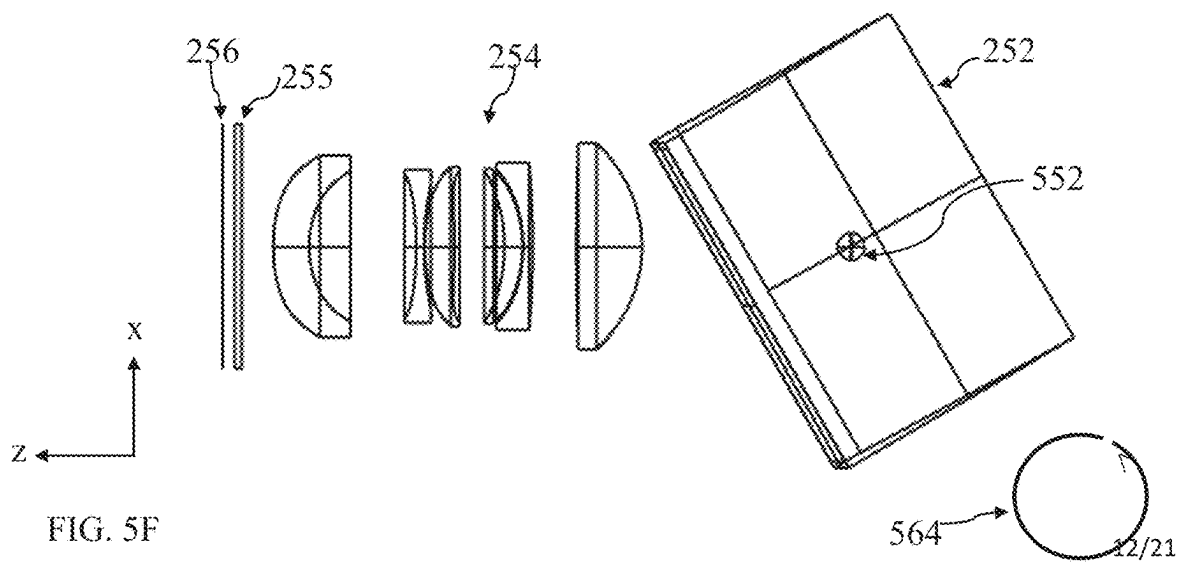
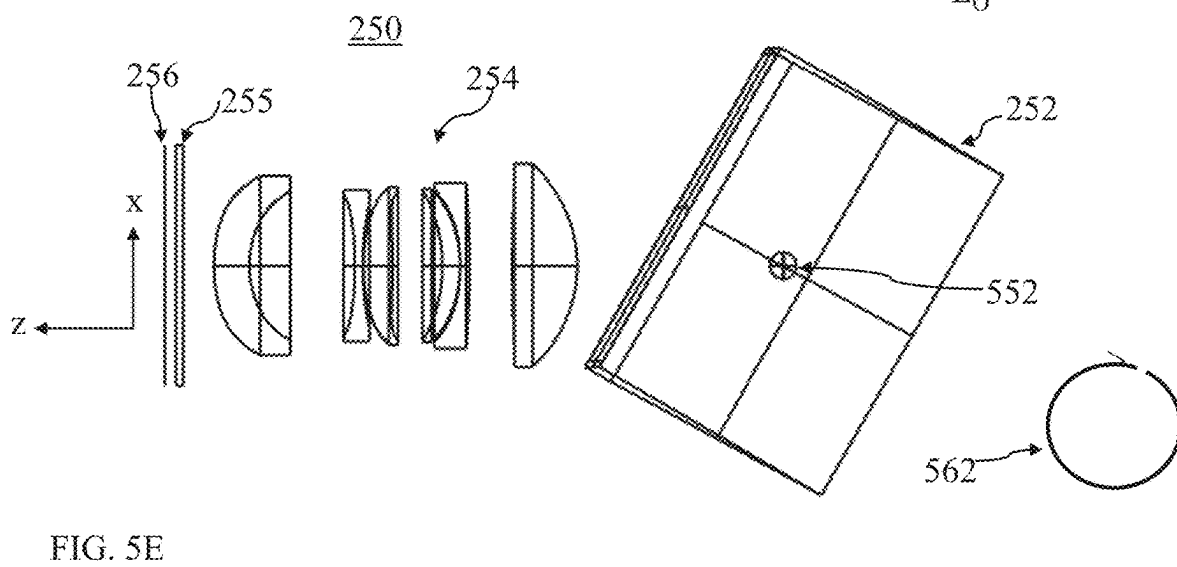
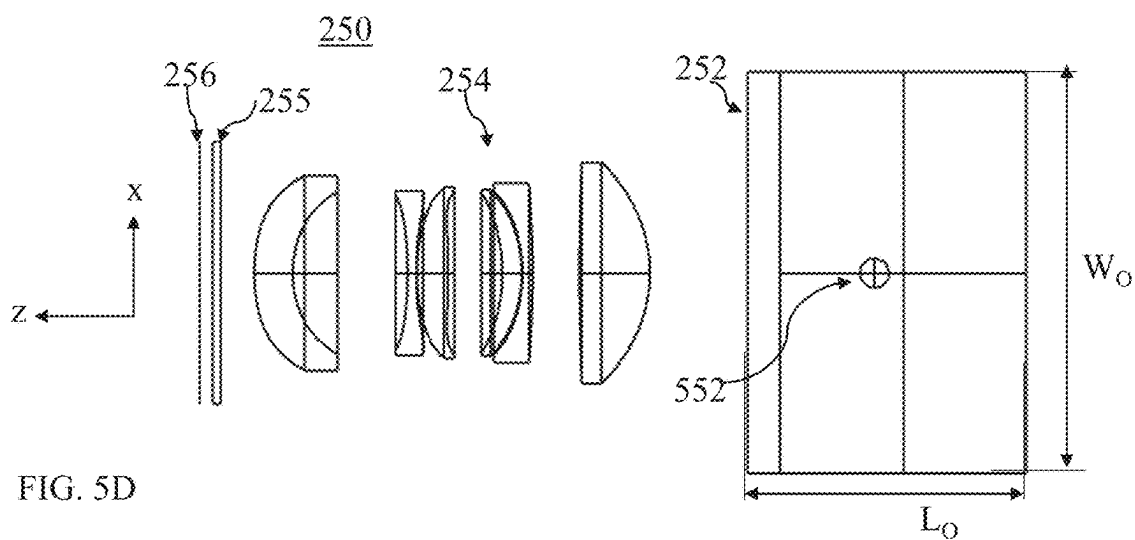


FIG. 5C



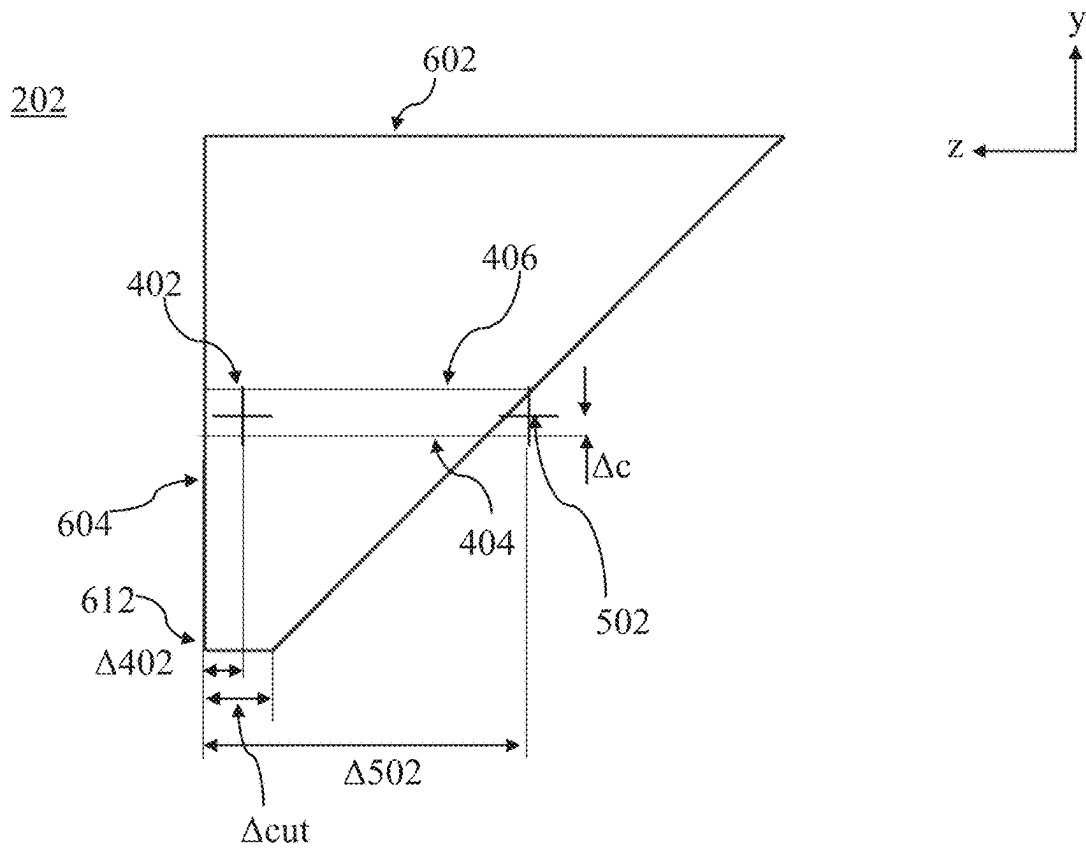


FIG. 6A

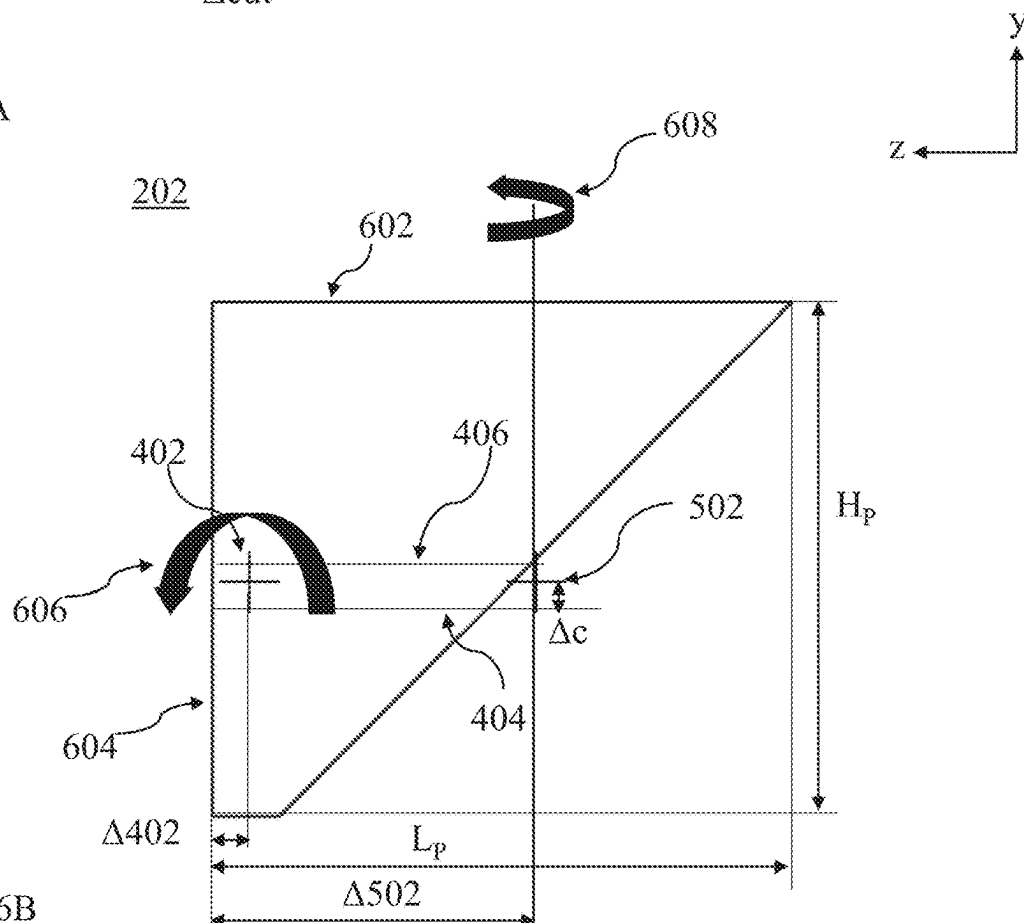


FIG. 6B

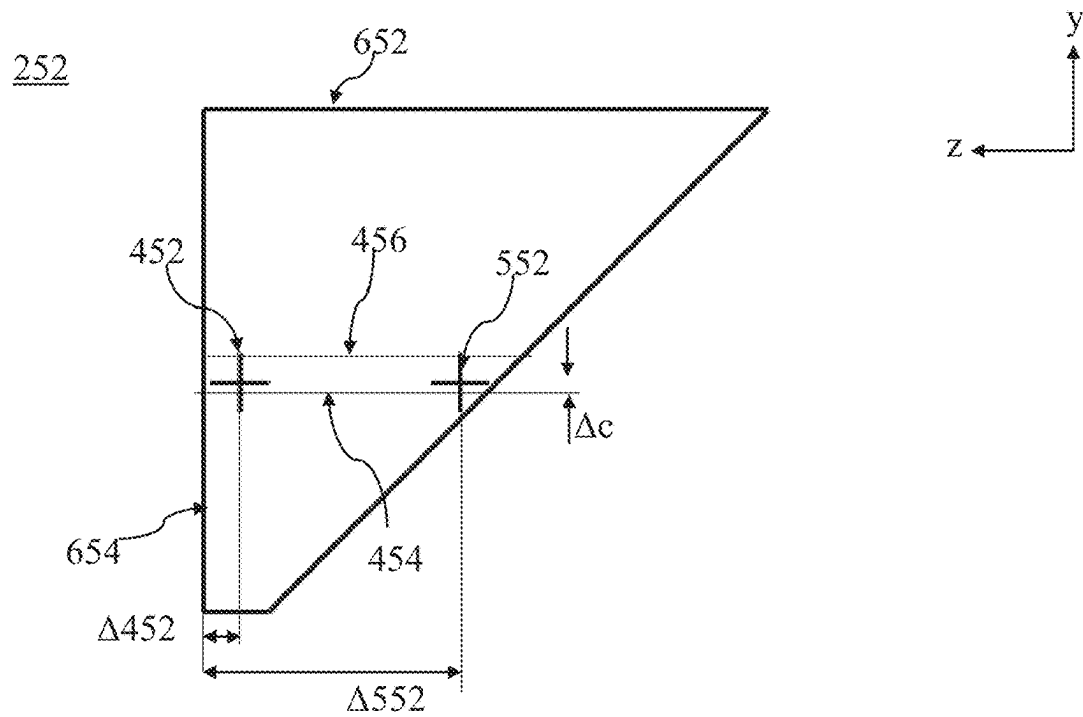


FIG. 6C

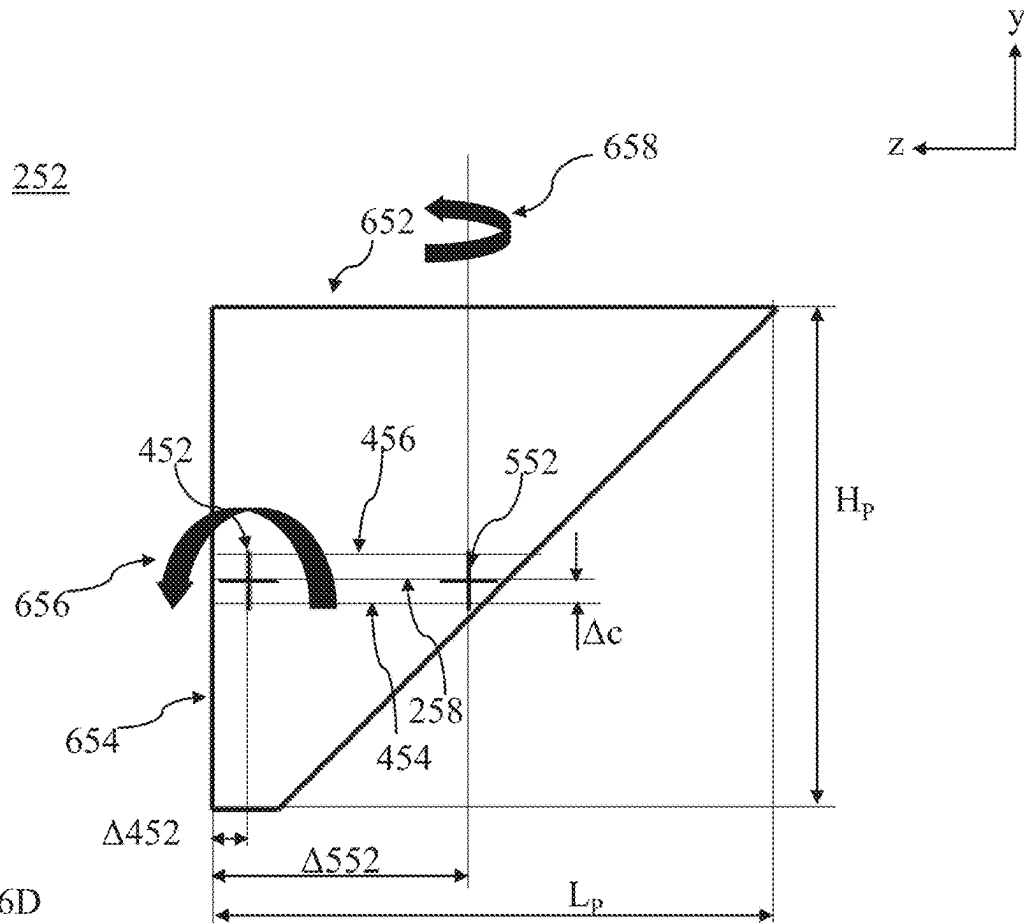


FIG. 6D

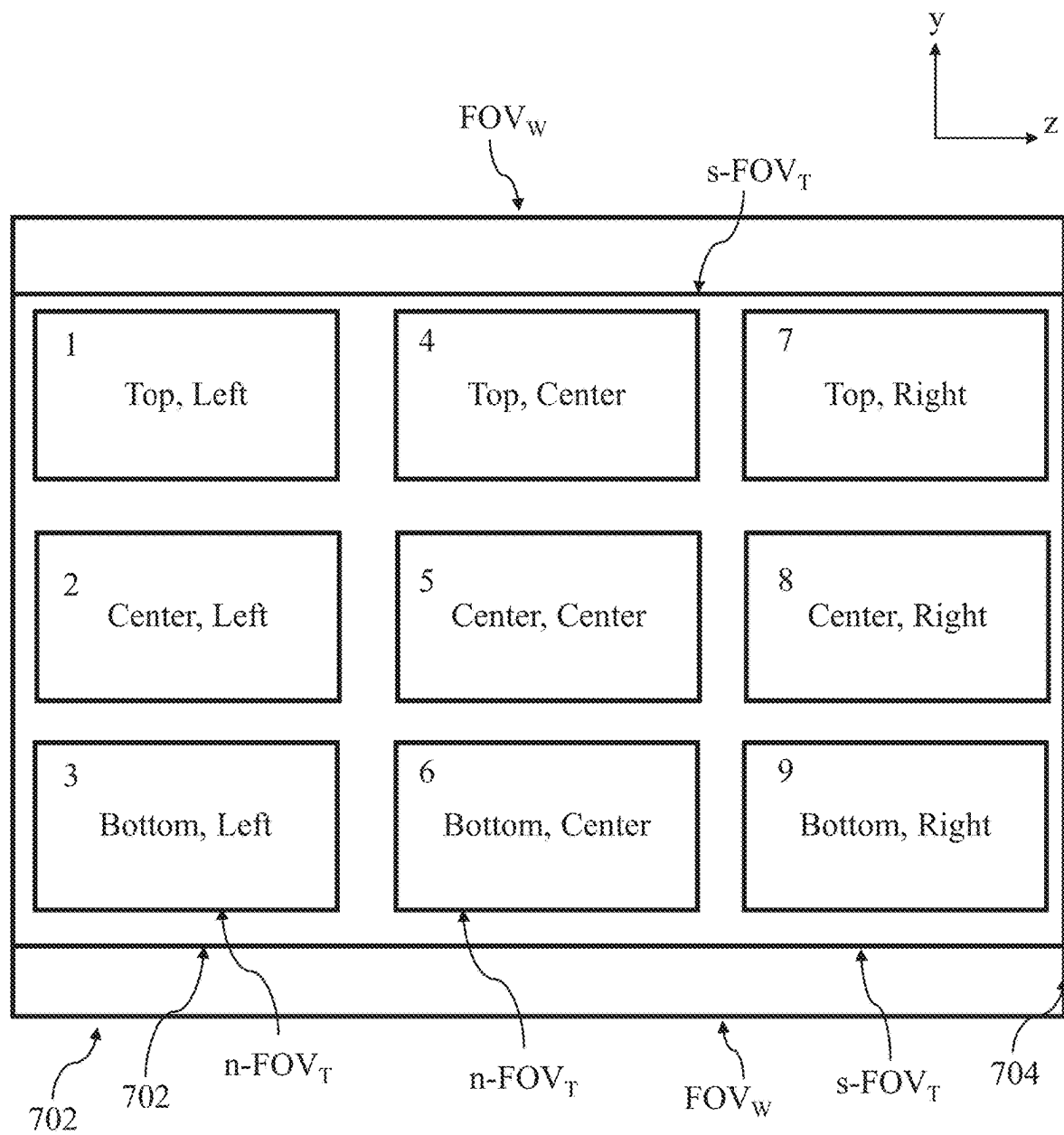


FIG. 7

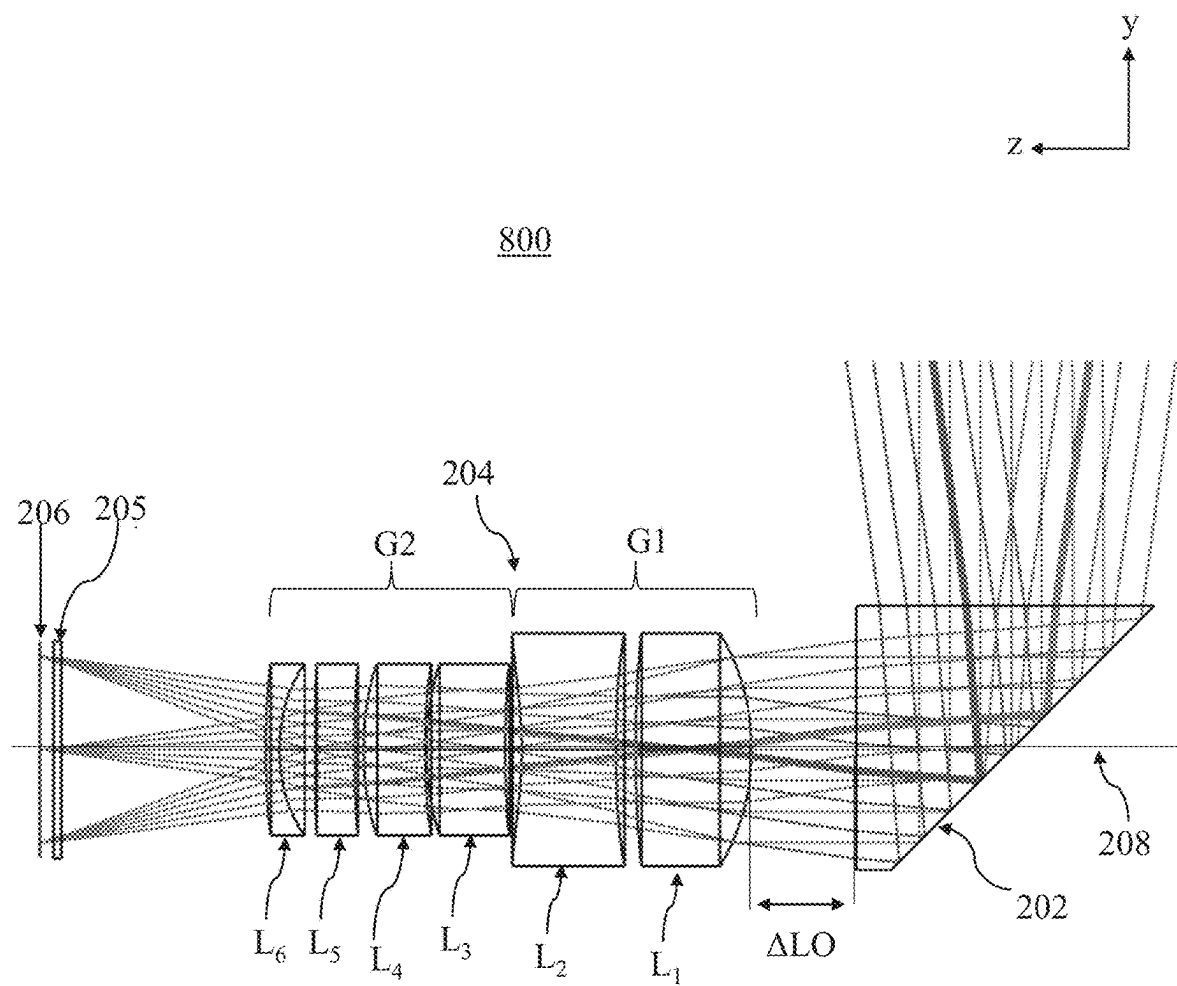


FIG. 8

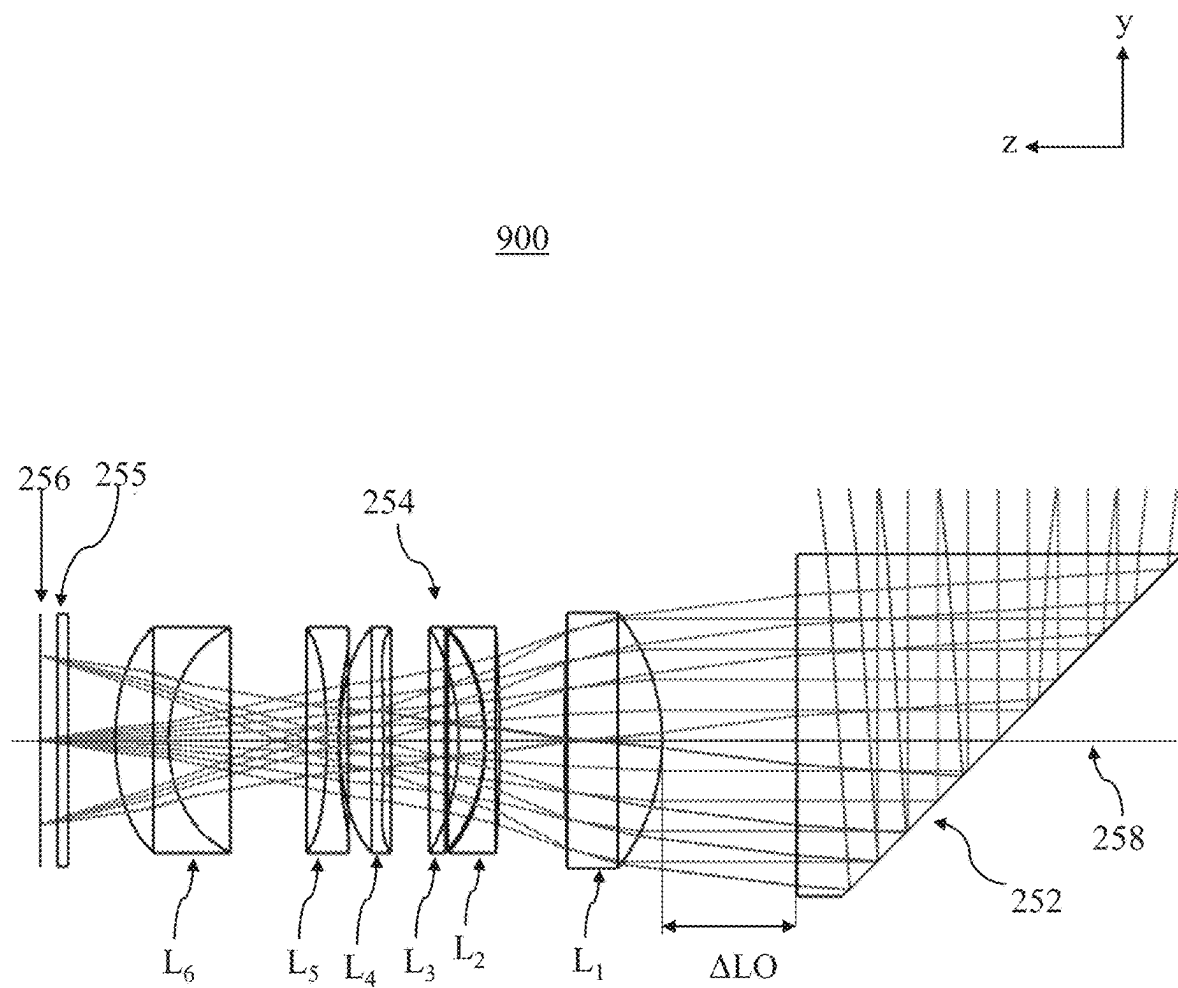


FIG. 9

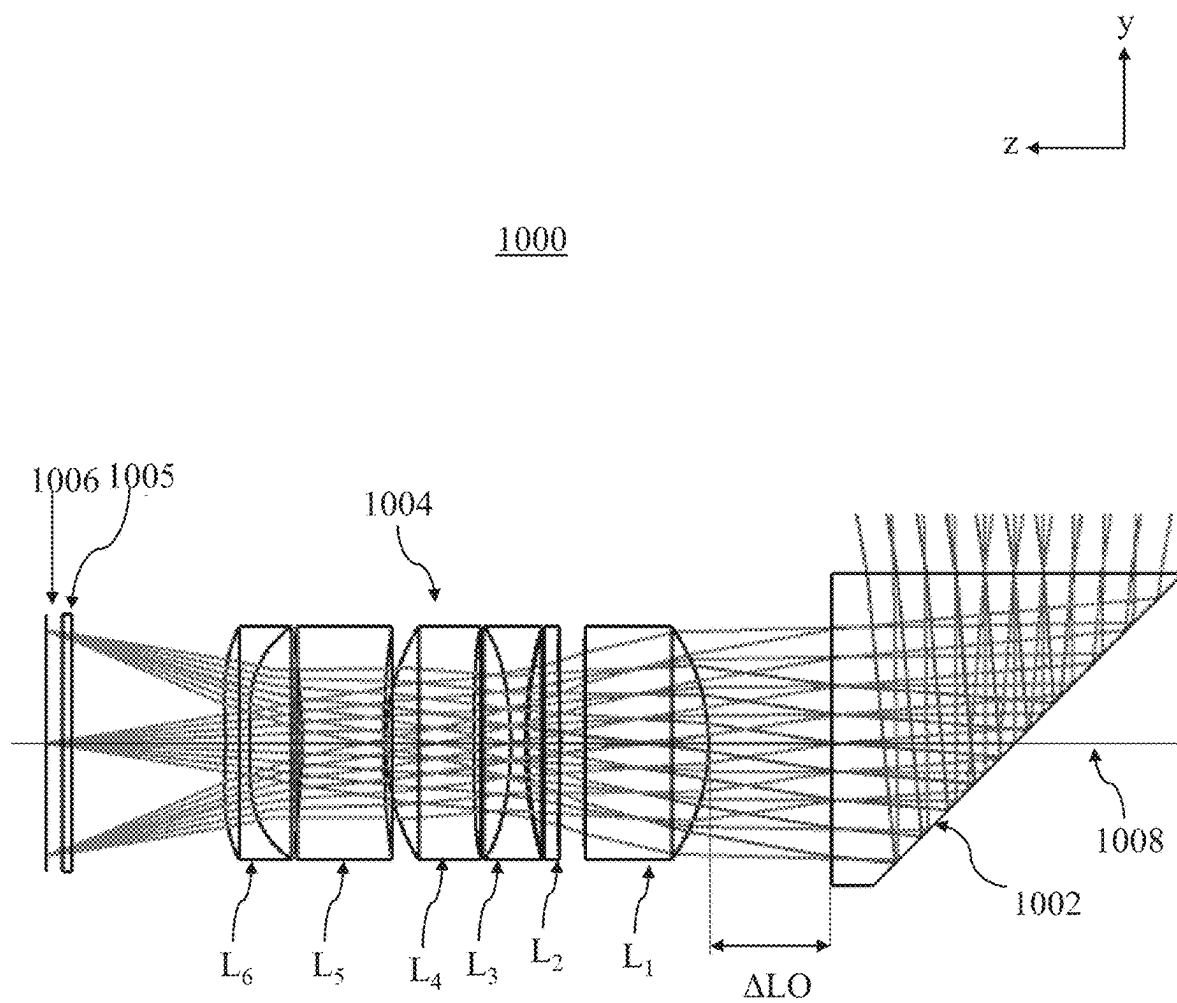


FIG. 10

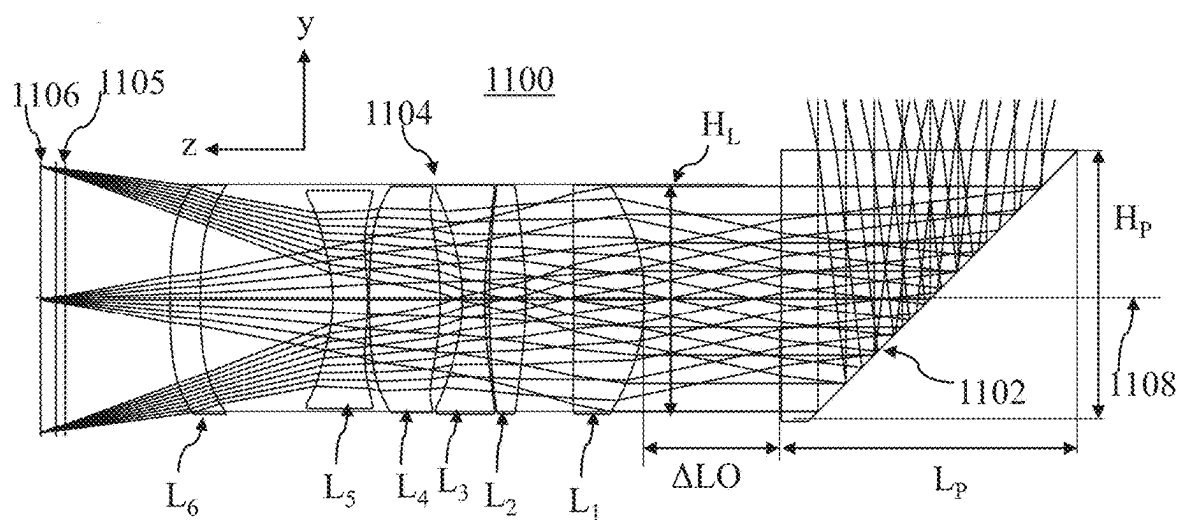


FIG. 11A

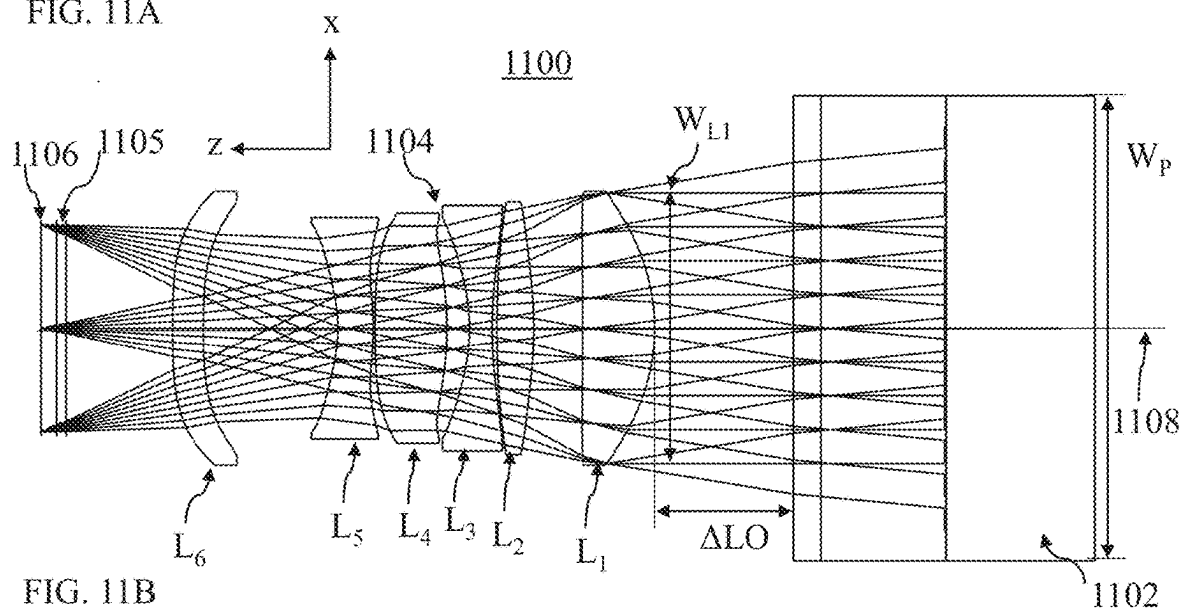


FIG. 11B

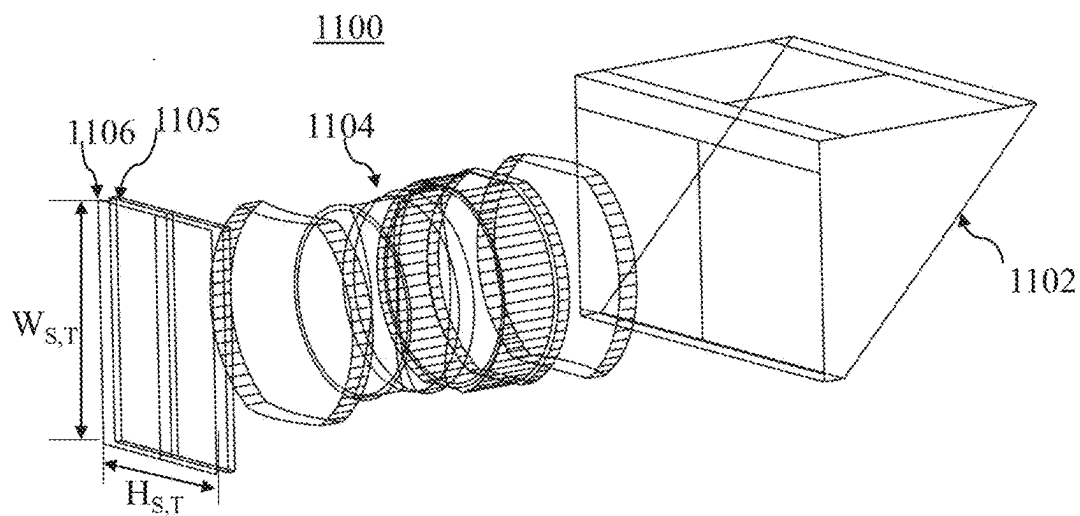


FIG. 11C

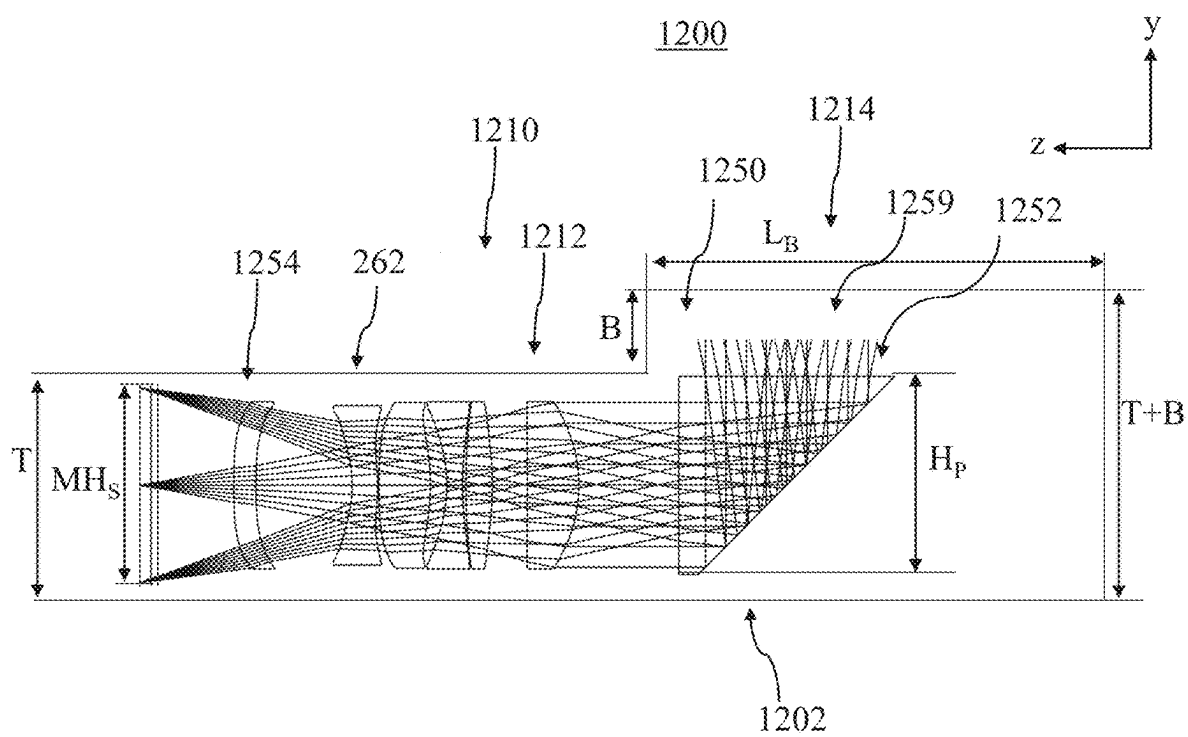


FIG. 12

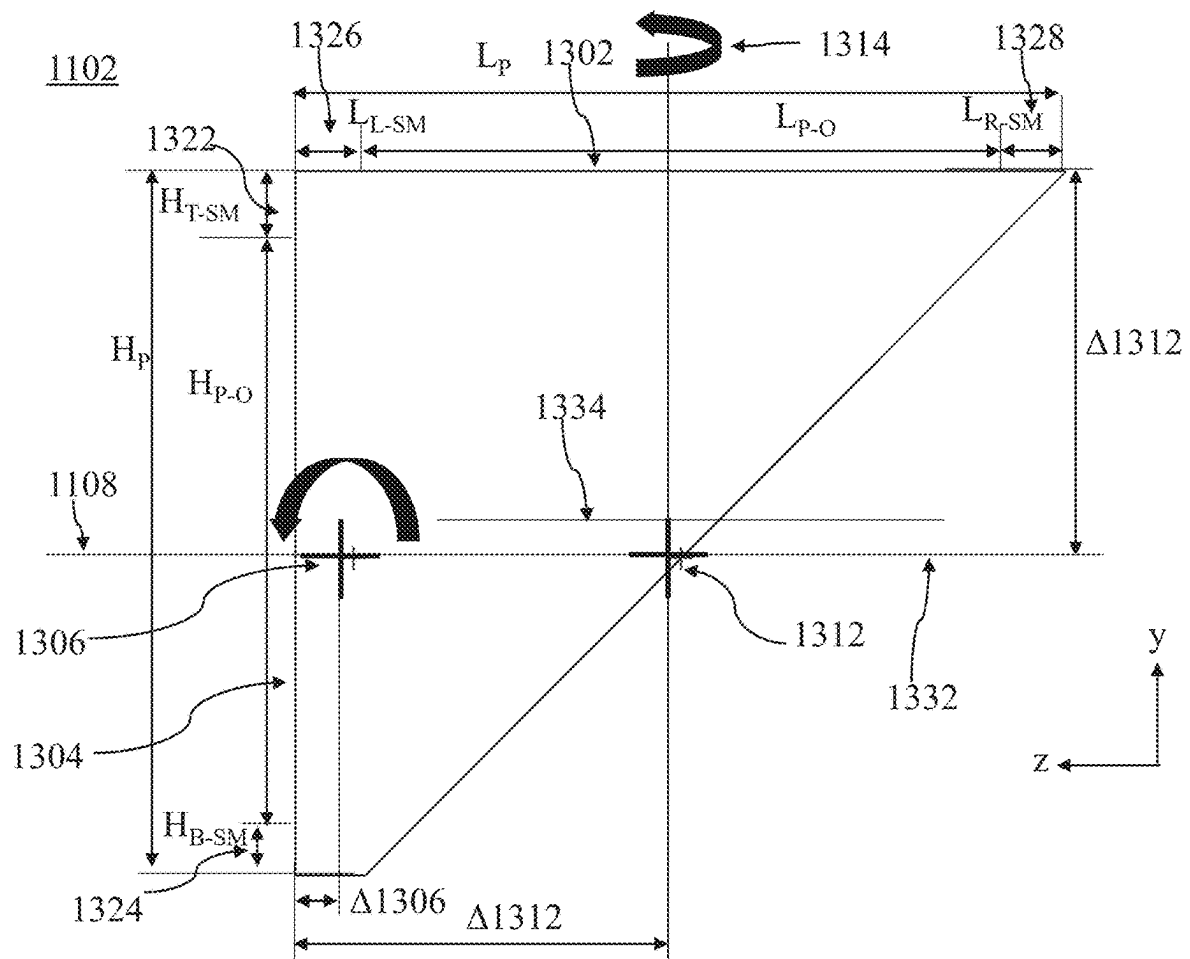


FIG. 13A

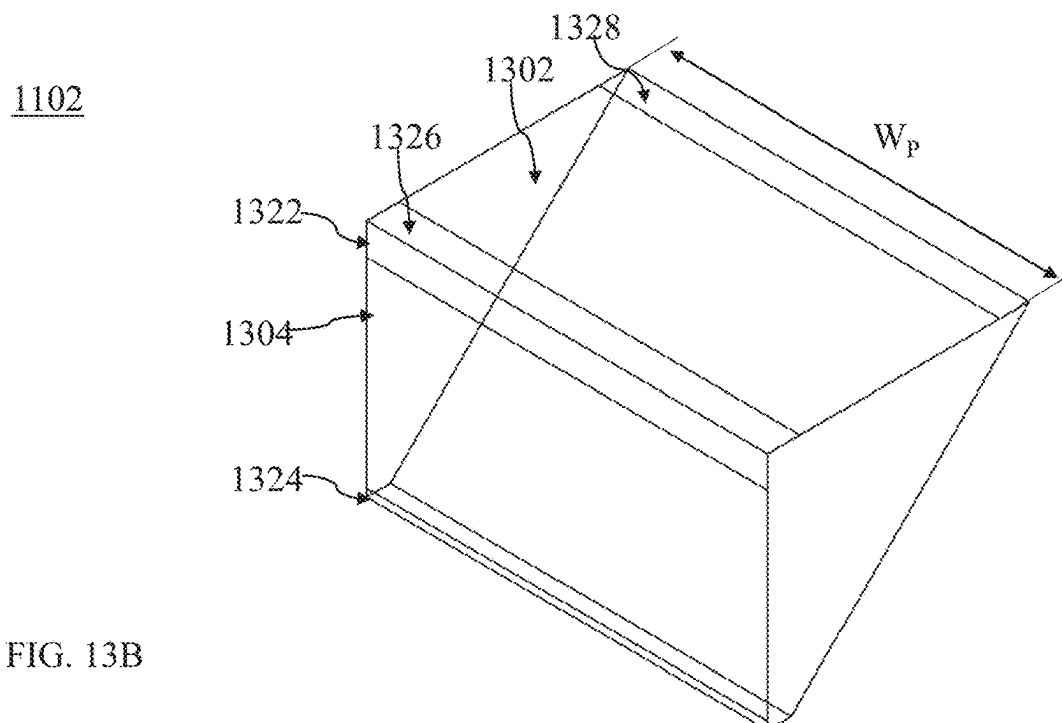


FIG. 13B

LARGE-APERTURE COMPACT SCANNING TELE CAMERAS

CROSS REFERENCE TO RELATED APPLICATIONS

This is a continuation from U.S. patent application Ser. No. 19/000,742 filed Dec. 24, 2024 (now allowed), which was a continuation from U.S. patent application Ser. No. 18/718,461 filed Jun. 11, 2024 (issued as U.S. Pat. No. 12,216,259), which was a 371 application from international patent application PCT/IB2022/060748 filed Nov. 8, 2022, which claims the benefit of priority from U.S. Provisional patent applications Nos. 63/289,323 filed Dec. 14, 2021, 63/297,256 filed Jan. 7, 2022, and 63/380,786 filed Oct. 25, 2022, all of which are incorporated herein by reference in their entirety.

FIELD

The subject matter disclosed herein relates in general to compact mobile cameras and in particular to mobile scanning telephoto (“Tele”) cameras.

DEFINITIONS

In this application and for optical and other properties mentioned throughout the description and figures, the following symbols and abbreviations are used, all for terms known in the art:

Lens element: a single lens element.

Lens: assembly of a plurality of lens elements.

Total track length (TTL): the maximal distance, measured along an axis parallel to the optical axis of a lens, between a point of the front surface S_1 of a first lens element L_1 and an image sensor, when the system is focused to an infinity object distance.

Back focal length (BFL): the minimal distance, measured along an axis parallel to the optical axis of a lens, between a point of the rear surface S_{2N} of the last lens element L_N and an image sensor, when the system is focused to an infinity object distance.

Effective focal length (EFL): in a lens (assembly of lens elements L_1 to L_N), the distance between a rear principal point P' and a rear focal point F' of the lens.

f-number ($f/\#$): the ratio of the EFL to an entrance pupil (or aperture) diameter of a lens.

BACKGROUND

Mobile electronic handheld devices (or just “mobile devices”) such as smartphones having two or more compact cameras (also referred to as “multi-cameras”) are known. The two or more cameras have lenses with different effective focal lengths (EFLs) that capture images of a same scene with different fields of view (FOVs). For example, a multi-camera may include a Wide camera having a Wide camera FOV (“FOV_W”) of e.g. 80 degrees and a Tele (or “zoom”) camera having a narrower FOV (“native FOV_T” or “n-FOV_T”) of e.g. 25 degrees and with higher spatial resolution (for example 3-5 times higher) than that of the Wide camera.

Tele cameras with scanning capability (“scanning Tele cameras” or “STCs”) for expanding the native fields-of-view n-FOV_T to an effective Tele FOV (also referred to as “scanning FOV_T” or “s-FOV_T”) overcome some of the limitations that relate to narrow n-FOV_Ts. Compact STCs

can be realized in a folded camera such as described for example in co-owned U.S. Pat. No. 10,578,948, or in a double-folded camera such as described for example in co-owned international patent PCT/IB2021/059843. One or two optical path folding elements (OPFEs), for example prisms or mirrors, are rotated along one or two directions to direct (or “scan” or “steer”) the n-FOV_T towards arbitrary points of view (POVs) within s-FOV_T.

The f-number (“ $f/\#$ ”) of a camera lens is the ratio of the EFL to the aperture diameter (“DA”): $f/\# = \text{EFL}/\text{DA}$. As known in the art, a low $f/\#$ is desired as of 3 major advantages: High signal-to-noise ratio (SNR), strong “natural” Bokeh effect and support of high image resolutions. A low $f/\#$ is, amongst others, achieved by maximizing the aperture area of the camera lens.

It is noted that herein, “aperture” refers to an entrance pupil of a lens (or “lens assembly”). If it is referred to an “aperture of a camera” or an “aperture of an optical lens system”, this always refers to the aperture of the lens included in the camera or in the optical lens system respectively. “Aperture” and “clear aperture” are used interchangeably.

There is need and it would be beneficial to have a compact scanning Tele camera for incorporation in a mobile device that includes a lens having a large aperture area and low f number.

SUMMARY

In various exemplary embodiments (examples), there is provided a scanning Tele cameras (STC) comprising: an OPFE for folding a first optical path OP1 to a second optical path OP2, wherein the OPFE has an OPFE height H_O measured along OP1, an OPFE length L_O measured along an axis parallel to OP, and a light exiting surface; an OPFE actuator; a lens having a lens optical axis parallel to OP2, an EFL, a maximum lens aperture height H_A measured along OP1, and a maximum lens aperture width W_A measured along an axis perpendicular to both OP1 and OP2; and an image sensor having an image sensor diagonal (SD) and an image sensor height H_{sensor} measured along OP1, wherein the STC has a STC native field-of-view (n-FOV_T), wherein the OPFE actuator is configured to rotate the OPFE around a first rotation axis perpendicular to both OP1 and OP2 and around a second rotation axis parallel to OP1 for scanning a scene with the n-FOV_T, wherein the first rotation axis is located at a distance $\Delta 1$ from the light exiting surface of the OPFE, and wherein $\Delta 1/L_O < 0.25$.

In some examples, $\Delta 1/L_O < 0.2$. In some examples, $\Delta 1/L_O < 0.15$. In some examples, $\Delta 1/L_O < 0.1$. In some examples, $\Delta 1/L_O < 0.075$.

In some examples, the OPFE has an OPFE center with respect to OP1, wherein the first rotation axis is located at a distance Δ_C from the OPFE center along OP1, and wherein a ratio of Δ_C and H_O fulfils $\Delta_C/H_O > 0.015$. In some examples, $\Delta_C/H_O > 0.02$.

In some examples, the lens is movable for focusing along OP2.

In some examples, the STC is included in a camera module, wherein the camera module is divided into a module region having a module region height H_M and a shoulder region having a shoulder region height $H_S < H_M$, all heights being measured along OP1, and wherein $H_S < H_A + 3$ mm. In some examples, the OPFE is included in the module region and the lens and the image sensor are included in the shoulder region.

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In some examples, the lens is divided into a first lens group (G1) and a second lens group (G2), wherein the OPFE and G1 are included in the module region and G2 and the image sensor are included in the shoulder region.

In some examples, the OPFE has an OPFE center with respect to OP1, wherein the first rotation axis is located at a distance Δ_C from the OPFE center along OP1, and a ratio of Δ_C and H_S fulfils $\Delta_C/H_S > 0.01$. In some examples, $\Delta_C/H_S > 0.015$.

In some examples, $H_S < H_A + 2$ mm.

In some examples, $H_S < W_A$.

In some examples, $H_S/H_M < 0.9$. In some examples, $H_S/H_M \leq 0.8$.

In some examples, $H_A/H_S > 0.7$.

In some examples, $H_A/H_M > 0.5$.

In some examples, $DA/H_S > 0.8$. In some examples, $DA/H_M > 0.65$.

In some examples, $H_M < H_O + 4$ mm. In some examples, $H_M < H_O + 3$ mm.

In some examples, the STC is included in a mobile device, wherein the mobile device has a regular region with a regular thickness T and a bump region with a bump thickness T+B, wherein the shoulder region is included in the mobile device regular region and wherein the module region is included in the mobile device bump region. In some examples, the mobile device may additionally include a Wide camera having a Wide camera image sensor and a Wide camera field-of-view (FOV_W). In some examples, the mobile device may be a smartphone.

In some examples, $H_O < H_A + 2$ mm. In some examples, $H_O < H_A + 1$ mm.

In some examples, $SD/EFL > 0.4$ mm.

In some examples, the STC uses a parallel STC sensor configuration. In some examples, the STC uses an anti-parallel STC sensor configuration.

In some examples, the scanning provides an effective Tele scanning FOV s-FOV_T, wherein the s-FOV_T has a longer horizontal side and a shorter vertical side, and wherein a horizontal side H-FOV_T of s-FOV_T is greater than 40 degrees.

In some examples, a center location of s-FOV_T is identical with a center location of FOV_W. In some examples, FOV_W is in the range of 50-120 degrees, wherein s-FOV_T covers a 16:9 segment of FOV_W. In some examples, FOV_W is in the range of 70-90 degrees, wherein s-FOV_T covers a 16:9 segment of FOV_W. In some examples, FOV_W is in the range of 75-85 degrees, wherein s-FOV_T covers a 16:9 segment of FOV_W.

In some examples, H-FOV_T > 45 degrees. In some examples, H-FOV_T > 50 degrees.

In some examples, s-FOV_T has a longer horizontal side and a shorter vertical side, and a vertical side V-FOV_T of s-FOV_T is greater than 20 degrees. In some examples, V-FOV_T > 25 degrees. In some examples, V-FOV_T > 30 degrees.

In some examples, the rotation of the OPFE along the first rotation axis is by more than ± 5 degrees around a zero scan position. In some examples, the rotation of the OPFE along the second rotation axis is by more than ± 15 degrees around a zero scan position.

In some examples, the OPFE is a prism. In some examples in which the OPFE is a prism, the prism has a fast scanning axis and a slow scanning axis, wherein the image sensor is oriented such that the prism's fast scanning axis is aligned with the horizontal side H-FOV_T of S-FOV_T.

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In some examples, the STC has an EFL of 8-10 mm. In some examples, the STC has an EFL of 10-25 mm. In some examples, the STC has an EFL of 25-50 mm.

In some examples, the OPFE actuator is a voice coil motor. In some examples, the lens has a f number f/#, wherein $f/\# < 3.5$. In some examples, $f/\# < 3$. In some examples, $f/\# < 2.5$. In some examples, a distance between the OPFE and the lens is ΔL_O , and $\Delta L_O/TTL < 0.25$.

In some examples, $H_O/L_O < 0.9$. In some examples, $W_O/H_O > 1.5$. In some examples, $W_O/H_O > 1.75$.

In some examples, the lens is a cut lens and the cut is performed along an axis parallel to OP2. In some examples, the cut lens is cut by 10%-50%. In some examples, the cut lens is cut by X %, wherein the cutting by X % reduces MH_M and MH_S by 0.5-X %-X %.

In some examples, the OPFE is a cut OPFE and the cut is performed along an axis parallel to OP2. In some examples, the cut OPFE is cut by 10%-40%. In some examples, the cut OPFE has a cut surface that extends along a distance Δcut from the OPFE light exiting surface, wherein $\Delta cut > 41$.

In some examples, the lens elements in the lens have an average lens thickness (ALT), the thickness of the first lens element L₁ being T1, and $T1/ALT > 1.5$.

In some examples, the lens elements in G1 have an ALT of ALT_{G1}, and $ALT_{G1}/ALT > 1.25$.

In some examples, the lens elements in G2 have an ALT of ALT_{G2}, and $ALT_{G1}/ALT_{G2} > 2$.

In some examples, the focal length of the first lens element is f1 and $f1/EFL < 0.75$. In some examples, a ratio of the height of G1 and the height of G2 is $H_{G1}/H_{G2} > 1.15$.

In some examples, $H_{G1}/H_{G2} > 1.3$.

In some examples, the first lens element L1 is made of glass.

In some examples in which the OPFE is a prism, the prism includes a stray light prevention mechanism. In some examples, the stray light prevention mechanism includes two stray light masks located at the light entering surface and two stray light masks located at the light exiting surface. In some examples, the two stray light masks located at the light entering surface are located at a left margin and at a right margin of the light entering surface, and wherein the two stray light masks located at the light exiting surface are located at a top and at a bottom of the light exiting surface. In some examples, the two stray light masks located at the light entering surface together cover a surface area of more than 10% and less than 20% of the area of the light entering surface. In some examples, the two stray light masks located at the light exiting surface together cover a surface area of more than 20% and less than 30% of the area of the light exiting surface.

In some examples there is provided a mobile device comprising an STC as above or below, together with an application processor (AP). In some examples, in a mobile device that includes a Wide camera, the AP is configured to use image data from the Wide camera for autonomous scanning of a scene with the STC's n-FOV_T. In some examples, the AP is configured to scan a scene with the STC's n-FOV_T according to a user input.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting examples of embodiments disclosed herein are described below with reference to figures attached hereto that are listed following this paragraph. The drawings and descriptions are meant to illuminate and clarify embodiments disclosed herein, and should not be considered lim-

iting in any way. Like elements in different drawings may be indicated by like numerals. Elements in the drawings are not necessarily drawn to scale.

FIGS. 1A-1D show known dual cameras including a folded STC included in a mobile device;

FIG. 1E shows an OPFE of a known STC as in FIGS. 1A-1D in a zero scan position in a cross-sectional view;

FIG. 2A shows a folded STC disclosed herein in a cross-sectional view;

FIG. 2B shows a folded STC disclosed herein in a top view;

FIG. 2C shows another folded STC disclosed herein in a cross-sectional view;

FIG. 2D shows another folded STC disclosed herein in a top view;

FIG. 3A shows a mobile device including the folded STC from FIGS. 2A-B in a cross-sectional view;

FIG. 3B shows another mobile device including the folded STC from FIGS. 2C-D in a cross-sectional view;

FIG. 3C shows yet another mobile device including a folded camera of FIGS. 2A-D and a Wide camera in a perspective view;

FIG. 3D shows the mobile device of FIG. 3C in a top view;

FIG. 3E shows the mobile device of FIG. 3C in a side view;

FIG. 3F shows yet another mobile device including a STC disclosed herein as well as a Wide camera;

FIG. 3G shows the mobile device of FIG. 3F in a top view;

FIG. 3H shows the mobile device of FIG. 3F in a side view;

FIGS. 4A-4C show the folded STC from FIGS. 2A-B at different scan states around a first rotation axis in a cross-sectional view;

FIGS. 4D-4F show the folded STC from FIGS. 2C-D at different scan states around a first rotation axis in a cross-sectional view;

FIGS. 5A-5C show the folded STC from FIGS. 2A-B at different scan states around a second rotation axis in a cross-sectional view;

FIGS. 5D-5F show the folded STC from FIGS. 2C-D at different scan states around a second rotation axis in a cross-sectional view;

FIGS. 6A and 6B show OPFE of the folded STC from FIGS. 2A-B and the first and the second rotation axis;

FIGS. 6C and 6D show OPFE of the folded STC from FIGS. 2C-D and the first and the second rotation axis;

FIG. 7 shows a field-of-view of the folded STC from FIGS. 2A-B and FIGS. 2C-D;

FIG. 8 shows an optical lens system included in the folded STC from FIGS. 2A-B.

FIG. 9 shows an optical lens system included in the folded STC from FIGS. 2C-D.

FIG. 10 shows another optical lens system disclosed herein;

FIG. 11A shows yet another optical lens system disclosed herein in a cross-sectional view;

FIG. 11B shows yet another optical lens system disclosed herein in a top view;

FIG. 11C shows yet another optical lens system disclosed herein in a perspective view;

FIG. 12 shows a mobile device including a STC which includes the yet another optical lens system of FIGS. 11A-C;

FIG. 13A shows a prism included in the yet another optical lens system of FIGS. 11A-C in a side view;

FIG. 13B shows the prism of FIG. 13A in a perspective view;

DETAILED DESCRIPTION

FIG. 1A illustrates a known folded scanning Tele camera (“folded STC”) **100** comprising an optical path folding element (OPFE) **102**, a lens **104** including a plurality of lens elements (not shown), lens **104** being included in a lens barrel **110**, and an image sensor **106**. Lens **104** has an optical lens height H_L , measured along OP **112**. H_L defines an aperture diameter (DA) of lens **104** in the y-direction. OPFE **102** folds an optical path (OP) from a first OP **112** (parallel with the y-axis in the YZ coordinate system shown) to a second OP **108** parallel with an optical axis of lens **104** along the z axis in the coordinate system shown. A theoretical limit for a height of a camera module (“minimum module height” or “MH_M”) and a theoretical limit for a length of a camera module (“minimum module length” or “ML_M”) including camera **100** is shown. MH_M and ML_M are defined by the largest dimension along OP **112** and along OP **108** of a component included in camera **100** respectively. For scanning a scene with STC **100**’s n-FOV_T, OPFE **102** may be rotated around two axes, a first rotation axis being parallel to the y-axis, and a second rotation axis being parallel to the x-axis.

FIG. 1B illustrates a known dual-camera **150** that comprises STC **100** and a (vertical or upright) Wide camera **130** including a Wide lens **132** and a Wide image sensor **138**. Lens **132** is included in a lens barrel **134**. Wide camera **130** has an OP **136** which is substantially parallel with OP **112**.

FIG. 1C shows schematically a known mobile device **160** (e.g. a smartphone) having an exterior rear surface **162** and including known STC **100** in a cross-sectional view. The aperture of STC **100** is located at rear surface **162**, a front surface **166** may e.g. include a screen (not visible). Mobile device **160** has a regular region of thickness (“T”) and a camera bump region **164** that is elevated by a height B over the regular region. The bump region has a bump length (“L_B”) and a bump thickness T+B. As shown, STC **100** is entirely integrated in the bump region, such that MML and MMH define a lower limit for the bump region, i.e. for L_B and T+B. For industrial design reasons, a small camera bump (i.e. a short L_B) is desired. Mobile device **160** may additionally include an application processor (AP—not shown). In some examples, the AP may be configured to scan a scene with STC **100**’s n-FOV_T according to a user input. In other examples, the AP may be configured to use image data from a Wide camera such as camera **130** to autonomously scan a scene with STC **100**’s n-FOV_T.

FIG. 1D shows a known cut lens element **180** in a cross-sectional view. Lens element **180** is cut by 20%, i.e. **180**’s optical width W_L is 20% larger than its optical height H_L . This means that also the aperture changes accordingly, such that the aperture is not axial symmetric. The cutting allows for a small H_L , which is required for small MH_Ms (see FIG. 1A), and still relatively large effective aperture diameters (DAs) which satisfy DA> H_L .

FIG. 1E shows OPFE **102** of known STC **100** in a zero scan position in a cross-sectional view. Here, OPFE **102** is a prism having a light entering surface **190** and a light exiting surface **191**. In the zero scan position, OPFE **102** has a length L_O (measured along the z-axis) and a height H_O (measured along the y-axis). The location of a first rotation axis **192** for rotating OPFE **102** around an axis perpendicular to the shown y-z-coordinate system (and as indicated by arrow **193**) and a second rotation axis **194** for rotating OPFE **102** around an axis parallel to the axis (and as indicated by arrow **195**) are shown. A distance from first rotation axis **192** to OPFE **102**’s light exiting surface **191** is marked Δ192. A

distance from second rotation axis **194** to OPFE **102**'s light entering surface **190** is marked $\Delta 194$. In a known STC, both rotation axes are located in a center region of OPFE **102**, i.e. a ratio between $\Delta 192$ and L_O and between $\Delta 194$ and H_O respectively is about 0.5. Specifically, $\Delta 192/L_O$ and $\Delta 194/H_O$ in general satisfy $\Delta 192/L_O=0.3-0.7$ and $\Delta 194/H_O=0.3-0.7$.

In the following, a "first rotation axis" of a prism indicates the rotation axis that does neither intercept with the light entering surface nor with the light exiting surface of a prism and which is parallel to both the light entering surface nor with the light exiting surface of a prism, as for example first rotation axis **192**. A "second rotation axis" of a prism indicates the rotation axis that intercepts with the light entering surface a prism and which is parallel to the light exiting surface of a prism, as for example second rotation axis **194**. It is noted that the first rotation axis as defined above represents a "fast scan axis" (or "efficient scan axis") of a STC, as for each degree of rotational movement of a prism around the first rotation axis, s-FOV_T moves by two degrees. The second rotation axis as defined above represents a "slow scan axis" (or "inefficient scan axis") of a STC, as for each degree of rotational movement of a prism around the second rotation axis, s-FOV_T moves by one degree.

In all examples disclosed herein, the OPFE is a prism having a light entering surface, a light reflecting surface and a light exiting surface. Therefore, we may use "OPFE" and "prism" interchangeably. However, this is not limiting, and in other examples a mirror having a light entering surface may be used.

FIG. 2A shows a STC disclosed herein and numbered **200** in a cross-sectional view. STC **200** includes an OPFE **202** (e.g. a prism or a mirror), a lens **204** including N=6 lens elements L_1-L_6 , an (optional) optical filter **205** and an image sensor **206**. FIG. 2B shows STC **200** in a top view.

Lens **204** has an optical axis **208**. STC **200** has an aperture **209**. STC **200** includes a camera module housing **210**. Module housing **210** has a module region **214** having a module height ("H_M") as well as a module length $L_{M,1}$ and a shoulder region **212** having a shoulder height ("H_S") that is lower by ΔH than module region **214**, i.e. $H_M > H_S$, as well as a shoulder length $L_{M,2}$. Here and in the following, all widths ("W") are measured along an axis parallel to the x-axis, all heights ("H") are measured along an axis parallel to the y-axis, all lengths ("L") are measured along an axis parallel to the z-axis.

A theoretical limit for a module height of camera **200** is "minimum module height" (or "MH_M"). A theoretical limit for a shoulder height of camera **200** is "minimum shoulder height" (or "MH_S"). MH_M and MH_S respectively are defined by the largest height dimension of a component included in STC **200**. MH_M is defined by OPFE **202**'s height H_O plus an additional height required for rotating OPFE **202**, as shown. In all STCs disclosed herein, a relatively low MH_M is achieved by making the two following design choices:

1. By locating or positioning a first OPFE rotation axis such as **402**, **452** and **1306** (see FIGS. 13A-13B) relatively close to the light exiting surface of an OPFE, i.e. by relatively small ratios of $\Delta 402/L_O$, $\Delta 452/L_O$ and $\Delta 1306/L_P$ which are smaller than 0.25.

2. Using an OPFE such as OPFE **202**, OPFE **252** and OPFE **1102** that fulfill $H_O < L_O$.

MH_S is defined by image sensor **206**'s height ("H_{sensor}") plus an additional height required for rotating OPFE **202**. Small MH_M and MH_S are beneficial for incorporating in slim mobile devices such as smartphones and tablets.

To clarify, all camera modules and optical lens systems disclosed herein are beneficially for use in mobile devices such as smartphones, tablets etc.

For achieving realistic estimations, we calculate H_M and H_S respectively by adding an additional height penalty of 1.5 mm to MH_M or MH_S, i.e. $H_M = MH_M + 1.5$ mm and $H_S = MH_S + 1.5$ mm. The penalty accounts for movements that may be required for optical image stabilization (OIS), autofocus (AF) as well as housing, lens cover etc. Note that the value of 1.5 mm is exemplary and by no means limiting, and that the addition may vary between 1 and 3 mm.

Lens **204** is divided in two lens groups, a first lens group ("G1") including L1 and L2 and a second lens group ("G2") including L3-L6. G1 has a maximal optical lens height H_{G1} and G2 has a maximal optical lens height H_{G2} , wherein $H_{G1} > H_{G2}$. G1 may be included in the module region **214** and G2 may be included in the shoulder region **212**. G1 has a maximal optical lens width W_{G1} and G2 has a maximal optical lens width W_{G2} , wherein $W_{G1} > W_{G2}$ (FIG. 2B). Also OPFE **202** may be included in module region **214**, and optical filter **205** and image sensor **206** may be included in shoulder region **212**. In other embodiments, entire lens **204** (i.e. both G1 and G2) may be included in shoulder region **212**. This may be beneficial for integrating STC **200** in a slim mobile device, i.e. a mobile device having a low height. A large H_{G1} may be desired, as it allows STC **200** to have a large DA. A small H_{G2} may be desired, as it allows STC **200** to have a slim shoulder region **214**, i.e. small H_S.

For scanning a scanning Tele FOV ("s-FOV_T") with STC **200**'s native FOV_T ("n-FOV_T"), OPFE **202** is rotated along two dimensions. OPFE **202** is shown in several rotation states which are required for scanning s-FOV_T. The rotation for scanning s-FOV may be actuated by a voice coil motor (VCM). OPFE **202** is a cut (or "D-cut") prism.

FIG. 2C shows another STC disclosed herein and numbered **250** in a cross-sectional view.

FIG. 2D shows STC **250** in a top view. STC **250** includes an OPFE **252** (e.g. a prism or a mirror), a lens **254** including N=6 lens elements L_1-L_6 , an (optional) optical filter **255** and an image sensor **256**. OPFE **252** is shown in several rotation states which are required for scanning s-FOV_T. OPFE **252** is a cut (or "D-cut") prism.

Lens **254** is a cut lens. Lens **254** has an optical axis **258**, an optical lens height H_L and an optical lens width W_L . STC **250** has an aperture **259**. STC **250** includes a camera module housing **260**. Module housing **260** has a module region **264** with module height H_M as well as a module region length $L_{M,1}$ and a shoulder region **262** having shoulder height H_S that is lower by ΔH than H_M , i.e. $H_M > H_S$, as well as a shoulder region length $L_{M,2}$. For industrial design reasons, it is beneficial to minimize $L_{M,1}$, as it allows for mobile devices with a small L_B (FIG. 3B).

A theoretical limit for a module height and a shoulder height of STC **250** is MH_M and MH_S respectively, as defined above. H_M and H_S are calculated respectively by adding a penalty 1.5 mm to MH_M or MH_S, i.e. $H_M = MH_M + 1.5$ mm and $H_S = MH_S + 1.5$ mm.

Lens **254** is fully included in shoulder region **262**. OPFE **252** is included in module region **264**. Optical filter **255** and image sensor **256** are included in shoulder region **262**.

In other examples, one or more of the first lens elements may be included in module region **264**. For lens **254**, L₁, which has a larger height H_{L1} than all other lens elements, may be included in module region **264**.

FIG. 3A shows a mobile device **300** (e.g. a smartphone) including STC **200** from FIGS. 2A-B in a cross-sectional view. Mobile device **300** has a front surface **302** (e.g.

including a screen, not shown) and a rear surface 310 including STC 200's aperture 209. Mobile device 300 has a regular region 312 of thickness "T" and a camera bump region 314 that is elevated (protrudes outwardly) by a height B over regular region 312. The bump region has a bump length (L_B) and a bump thickness T+B. From an industrial design point of view, it is desired to minimize a bump area, i.e. to have a short L_B . For achieving short L_B , module region 214 of STC 200 is included in bump region 314 and shoulder region 212 of STC 200 is included in regular region 312. This means that OPFE 202 and G1 of lens 204 are included in bump region 314, and G2 of lens 204 and image sensor 206 are included in regular region 312.

Optionally, in some embodiments (also referred to as "examples"), parts of shoulder region 212 may also be included in bump region 314. In other embodiments, both G1 and G2 of lens 204, i.e. the entire lens 204, are included in bump region 314.

FIG. 3B shows a mobile device numbered 320 (e.g. a smartphone) including STC 250 from FIGS. 2C-D in a cross-sectional view. Mobile device 320 has a front surface 322 (e.g. including a screen, not shown) and a rear surface 330 including STC 250's aperture 259. Mobile device 320 has a regular region 332 of thickness "T" and a camera bump region 334 that is elevated by a height B over regular region 332 and has a bump region length L_B . For achieving short L_B , module region 264 of STC 250 is included in bump region 334 and shoulder region 262 of STC 250 is included in regular region 332. OPFE 252 is included in bump region 334 and lens 254 and image sensor 256 are included in regular region 332. Mobile devices 300 and 320 may additionally include a Wide camera such as Wide camera 130 providing Wide camera images and an application processor (AP). In some examples, the AP may be configured to use Wide camera images to analyze a scene, and, based on the scene analysis, to scan the scene with STC 200's and STC 250's n-FOV_T autonomously. In other examples, the AP may be configured to scan a scene with STC 200's and STC 250's n-FOV_T based on a user's input. A FOV of the Wide camera ("FOV_w") may be in the range of 50 degrees-120 degrees or more, e.g. 80 degrees. In a zero scan position, a center position of n-FOV_T coincides with a center position of FOV_w. A center position of s-FOV_T coincides with a center position of FOV_w. In some examples, s-FOV_T covers a 16:9 segment of FOV_w.

FIGS. 3C-E show another mobile device numbered 340 (e.g. a smartphone) including a STC such as STC 200 from FIGS. 2A-B or STC 250 from FIGS. 2C-D, as well as a Wide camera 344. In the following and exemplarily, we refer to STC 200 only. FIG. 3C shows mobile device 340 in a perspective view. FIG. 3D shows mobile device 340 in a top view. FIG. 3E shows mobile device 340 in a side view. Wide camera 344 has a Wide camera lens (not shown), a Wide camera aperture 346 and a Wide camera image sensor 348. Mobile device 340 has a front surface 342 (e.g. including a screen, not shown) and a rear surface 350 including the apertures of the STC and Wide camera 344. When considered in top view (FIG. 3D), mobile device 340 has a rectangular shape and a device width " W_D " measured along the x-axis as well as a device height " H_D " measured along the z-axis, as shown. A ratio of $W_D:H_D$ is in general different from 1:1, and may be 16:9, 19:9 or similar. That is, in general $W_D > H_D$. Both image sensor 206 and image sensor 348 have a rectangular shape. In the following, a respective width and height of an image sensor is defined as follows: a width of an image sensor represents (or indicates) a largest dimension of the image sensor, and a height of an image

sensor represents a second largest dimension of the image sensor. STC 200's image sensor 206 has a Tele sensor width (" $W_{S,T}$ ") measured along the x-axis and a Tele sensor height (" $H_{S,T}$ ") measured along the y-axis, as shown. Wide camera 344's image sensor 348 has a Wide sensor width (" $W_{S,w}$ ") measured along the x-axis and a Wide sensor height (" $H_{S,w}$ ") measured along the z-axis, as shown. In general, Wide camera 344 is integrated into mobile device 340 such that $W_{S,w}$ is substantially parallel to W_D . For both image sensors, a ratio of $W_S:H_S$ is in general different from 1:1, and may be 4:3, 16:9 or similar. That is, in general $W_S > H_S$. Image sensor 206 and image sensor 348 may or may not have an identical ratio of $W_S:H_S$. In mobile device 340, STC 200 is integrated into mobile device 340 such that W_S is substantially parallel to both $W_{S,w}$ and W_D . $H_{S,T}$ is substantially perpendicular to $H_{S,w}$. When considering this incorporation of STC 200 into mobile device 340 in a side view, image sensor 206 is oriented parallel to mobile device 340 (i.e. the rectangular shape of image sensor 206 is oriented parallel to the rectangular shape of mobile device 340). Therefore, in the following we refer to this configuration as "parallel STC sensor configuration".

FIGS. 3F-H show yet another mobile device numbered 360 (e.g. a smartphone) including a STC 1250 as well as a Wide camera 344. STC 1250 includes an optical lens system such as optical lens system 1100 (see FIG. 11) including an OPFE 1102, a lens (not shown) and an image sensor 1106. FIG. 3F shows mobile device 360 in a perspective view. FIG. 3G shows mobile device 360 in a top view. FIG. 3H shows mobile device 360 in a side view. Mobile device 360 has a front surface 362 and a rear surface 370 including the apertures of STC 1250 and Wide camera 344. Mobile device 360 has a device width " W_D " measured along the x-axis as well as a device height " H_D " measured along the z-axis, as shown. Both image sensor 1106 and image sensor 348 have a rectangular shape. STC 1250's image sensor 1106 has a Tele sensor width (" $W_{S,T}$ ") measured along the y-axis and a Tele sensor height (" $H_{S,T}$ ") measured along the x-axis, as shown. Wide camera 344's image sensor 348 has a Wide sensor width (" $W_{S,w}$ ") measured along the x-axis and a Wide sensor height (" $H_{S,w}$ ") measured along the z-axis, as shown. In general, Wide camera 344 is integrated into mobile device 340 such that $W_{S,w}$ is substantially parallel to W_D . For both image sensors, a ratio of $W_S:H_S$ is in general different from 1:1, and may be 4:3, 16:9 or similar. Image sensor 1106 and image sensor 348 may or may not have an identical ratio of $W_S:H_S$. In mobile device 360, STC 1250 is integrated into mobile device 360 such that $W_{S,T}$ is substantially perpendicular to both $W_{S,w}$ and W_D , which are parallel to each other. $H_{S,T}$ is substantially perpendicular to $H_{S,w}$ substantially parallel to $W_{S,w}$. When considering this incorporation of STC 1300 into mobile device 360 in a side view, image sensor 1106 is oriented anti-parallel to mobile device 360 (i.e. the rectangular shape of image sensor 1106 is oriented anti-parallel to the rectangular shape of mobile device 360). Therefore, in the following we refer to this configuration as "anti-parallel STC sensor configuration".

FIG. 4A shows STC 200 from FIGS. 2A-B without module housing 210 in a zero scan position in a cross-sectional view. The zero scan position is defined by OPFE 202's top surface being parallel with the z-axis such that n-FOV_T is located at the center of s-FOV_T. OPFE 202 is rotated around a first rotation axis 402 which is parallel to the x-axis (i.e. perpendicular to both the y-axis and the z-axis shown). Non-cut center axis 404 indicates a center of a non-cut OPFE 202 with respect to the y-axis, i.e. if OPFE 202 was not cut, non-cut center axis 404 would be located

at the center of the non-cut OPFE. Cut center axis **406** indicates a center of cut OPFE **202** with respect to the y-axis, i.e. cut center axis **406** is located at the center of cut OPFE **202**. As visible, first rotation axis **402** intersects with the optical axis **208** of lens **204**. However, first rotation axis **402** does not intersect with non-cut center axis **404** nor with cut center axis **406**. First rotation axis **402** is located at distance ΔC from non-cut center axis **404**. This de-center location with respect to the y-axis of OPFE **202** is beneficial for minimizing MH_{Δ} . Image sensor **206** is oriented in a parallel STC sensor configuration.

FIG. **4B** shows STC **200** without module housing **210** from FIG. **4A** in a maximal counter-clockwise rotation state with respect to the first rotation axis **402**. The counter-clockwise rotation direction **412** is shown.

FIG. **4C** shows STC **200** without module housing **210** from FIGS. **4A-B** in a maximal clockwise rotation state with respect to the first rotation axis **402**. The clockwise rotation direction **414** is shown.

FIG. **4D** shows STC **250** from FIGS. **2C-D** without module housing **260** in a zero scan position in a cross-sectional view. OPFE **252** is rotated around a first rotation axis **452** which is parallel to the x-axis (i.e. perpendicular to both the y-axis and the z-axis shown). Non-cut center axis **454** indicates a center of a non-cut OPFE **252** with respect to the y-axis. Cut center axis **456** indicates a center of cut OPFE **252** with respect to the y-axis. First rotation axis **452** intersects with the optical axis **258** of lens **254**, but first rotation axis **452** does not intersect with non-cut center axis **454** nor with cut center axis **456**. First rotation axis **452** is located at distance ΔC from non-cut prism center axis **454**. This de-center location of OPFE **252** is beneficial for minimizing MH_{Δ} . Image sensor **256** is oriented in a parallel STC sensor configuration.

FIG. **4E** shows STC **250** without module housing **260** from FIG. **4D** in a maximal counter-clockwise rotation state with respect to the first rotation axis **452**.

FIG. **4F** shows STC **250** without module housing **260** from FIGS. **4D-E** in a maximal clockwise rotation state with respect to the first rotation axis **452**.

The counter-clockwise rotation direction **462** and the clockwise rotation direction **464** are shown.

FIG. **5A** shows STC **200** from FIGS. **2A-B** without module housing **210** in a zero scan position in a top view. The zero scan position is defined by OPFE **202**'s top surface being parallel with the z-axis such that n-FOV_T is located at the center of s-FOV_T. OPFE **202** is rotated around a second rotation axis **502** which is parallel to the y-axis (i.e. perpendicular to both the x-axis and the z-axis shown).

FIG. **5B** shows STC **200** without module housing **210** from FIG. **5A** in a maximal clockwise rotation state with respect to the second rotation axis **502**. The clockwise rotation direction **512** is shown.

FIG. **5C** shows STC **200** without module housing **210** from FIGS. **5A-B** in a maximal counter-clockwise rotation state with respect to the second rotation axis **502**. The counter-clockwise rotation direction **514** is shown.

FIG. **5D** shows STC **250** from FIGS. **2C-D** without module housing **260** in a zero scan position in a top view. OPFE **252** is rotated around a second rotation axis **552** which is parallel to the y-axis (i.e. perpendicular to both the x-axis and the z-axis shown).

FIG. **5E** shows STC **250** without module housing **260** from FIG. **5D** in a maximal clockwise rotation state with respect to the second rotation axis **552**.

FIG. **5F** shows STC **250** without module housing **210** from FIGS. **5D-E** in a maximal counter-clockwise rotation state with respect to the second rotation axis **552**.

The clockwise rotation direction **562** and the counter-clockwise rotation direction **564** are shown. FIG. **6A** shows OPFE **202** of STC **200** in a zero scan position in a cross-sectional view. The location of first rotation axis **402** and second rotation axis **502** are shown. A distance from first rotation axis **402** to OPFE **202**'s light exiting surface **604** is marked $\Delta 402$. Here, $\Delta 402=0.5$ mm. OPFE **202** is a cut prism which has a cut bottom corner (or margin) **612**. Considering the cutting along the z-axis, a cut surface extends from light exiting surface **604** along a distance Δ_{cut} , as shown. Beneficially, first rotation axis **402** is located within this distance Δ_{cut} . This is valid also for all other OPFEs disclosed herein such as OPFE **202**, OPFE **252** and OPFE **1102**.

A distance from second rotation axis **502** to OPFE **202**'s light entering surface **602** is marked $\Delta 502$. Here, $\Delta 502=4.3$ mm.

FIG. **6B** shows OPFE **202** of STC **200** from FIG. **6A** in a cross-sectional view. A first rotation direction **606** around first rotation axis **402** and a second rotation direction **608** around second rotation axis **404** are shown. A prism height H_P and a prism length L_P OPFE **202** are shown as well. Here, $L_P=7.1$ mm and $H_P=6.9$ mm. Ratios of $\Delta 402/L_P=0.07$ and $\Delta 502/H_P=0.62$.

FIG. **6C** shows OPFE **252** of STC **250** in a zero scan position in a cross-sectional view. OPFE **252** is a prism. The location of first rotation axis **452** and second rotation axis **552** are shown. A distance from first rotation axis **452** to OPFE **252**'s light exiting surface **654** is $\Delta 452$. Here, $\Delta 452=0.5$ mm. A distance from second rotation axis **552** to OPFE **252**'s light entering surface **652** is $\Delta 552$. Here, $\Delta 552=3.5$ mm.

FIG. **6D** shows OPFE **252** of STC **250** from FIG. **6C** in a cross-sectional view. A first rotation direction **656** around first rotation axis **452** and a second rotation direction **658** around second rotation axis **454** as well as a length L_P and height H_P of OPFE **252** are shown. Here, $L_P=7.2$ mm and $H_P=6.7$ mm. Ratios of $\Delta 452/L_P=0.07$ and $\Delta 552/H_P=0.52$.

FIG. **7** shows exemplary a Wide camera FOV ("FOV_W"), a s-FOV_T and 9 n-FOV_Ts (marked 1-9) of a known Wide camera and a STC such as STC **200** or STC **250** or STC **1250**. FOV_W shows a typical Wide camera FOV of for example 82° measured along a diagonal of FOV_W. In this example, a 16:9 FOV ratio of FOV_W covers about 69.4°×42.6° (i.e. 69° in a horizontal direction, 42° in a vertical direction). s-FOV_T shows a segment of a scene that can be covered with the STC, i.e. it includes all POVs that can be reached with the STC. In some examples, s-FOV_T may cover a 16:9 FOV ratio of FOV_W, as shown in FIG. **7**. As visible, FOV_W and s-FOV_T have a "longer side" **702** (here, along the z-axis) and a "shorter side" **704** (here, along the x-axis). s-FOV_T of STC **200** covers 50.9°×32.5° (50.9° in a horizontal direction, 32.5° in a vertical direction). The 9 n-FOV_Ts represent maximum scan positions. n-FOV_T 5, i.e. the (Center, Center) position, represents a zero scan position. For example, n-FOV_T 1 represents the n-FOV_T that is obtained when scanning STC **200** maximally to a top-left position, n-FOV_T 6 represents the n-FOV_T that is obtained when scanning STC **200** maximally to a bottom-center position etc. Table 1 provides the rotation values of OPFE **202** around (first rotation axis **402**, second rotation axis **502**) respectively that are required for scanning to the 9 respective n-FOV_Ts. The values refer to a scanning action that starts from n-FOV_T 5, i.e. the (Center, Center) position. For example for scanning n-FOV_T to n-FOV_T 9 or (Bottom,

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Right), starting from (Center, Center) position n-FOV_T 5, OPFE **202** must be rotated by -7.85 degrees around first rotation axis **402** and by -15.46 degrees around second rotation axis **502**.

TABLE 1

	Left	Center	Right
Top	(1.76, 21.67)	(4.86, 0)	(1.76, -21.67)
Center	(-3.16, 18.49)	(0, 0)	(-3.16, -18.49)
Bottom	(-7.85, 15.46)	(-4.86, 0)	(-7.85, -15.46)

For STC **250** including optical lens system **900**, Table 2 provides the rotation values of OPFE **252** around (first rotation axis **452**, second rotation axis **552**) respectively that are required for scanning to the 9 respective n-FOV_Ts shown in FIG. 7. s-FOV_T of STC **250** covers 69.5°×42.58°. This means that s-FOV_T of STC **250** covers a 16:9 ratio of a FOV_W having a diagonal FOV_W=82°, as shown in FIG. 7

TABLE 2

	Left	Center	Right
Top	(3.63, 29.80)	(8.35, 0)	(3.63, -29.80)
Center	(-5.80, 24.02)	(0, 0)	(-5.80, -24.02)
Bottom	(-11.67, 18.83)	(-8.35, 0)	(-11.67, -18.83)

For another STC (not shown) including optical lens system **1000**, Table 3 provides the rotation values of OPFE **1002** around a first rotation axis and around a second rotation axis respectively that are required for scanning to the 9 respective n-FOV_Ts shown in FIG. 7. s-FOV of the STC including optical lens system **1000** covers 69.5°×42.58°. This means that the s-FOV-covers a 16:9 ratio of a FOV_W having a diagonal FOV_W=82°.

TABLE 3

	Left	Center	Right
Top	(1.49, 25.54)	(5.79, 0)	(1.49, -25.54)
Center	(-4.31, 21.34)	(0, 0)	(-4.31, -21.34)
Bottom	(-9.67, 17.16)	(-5.79, 0)	(-9.67, -17.16)

In some examples, an OPFE may be rotated around one axis or around two axes for optical image stabilization (OIS). In some examples and per axis, an OPFE may be rotated by ±2 degrees or by ±5 degrees around a zero position for performing OIS. In other examples, an OPFE may be rotated by even ±10 degrees or more around a zero position for performing OIS. In these examples, in general a mobile device including the STC includes as well an additional sensor such as e.g. an inertial measurement unit (IMU) and a processor, e.g. an application processor (AP) or a micro controller unit (MCU). The additional sensor is used for sensing an undesired rotation of the mobile device, and based on the sensing data of the additional sensor, the

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processor calculates OPFE rotation control signals which control a rotational movement of the OPFE that mitigates (or counteracts) the undesired rotation of the mobile device.

FIG. 8 shows optical lens system **800** included in STC **200** from FIGS. 2A-B in a cross-sectional view and with ray-tracing. A distance ΔL_O between OPFE **202** and lens **204** is 2.7 mm.

The optical height (H_{L1}) and width (W_{L1}) of lens element L₁ may define the optical height and width of G1 (i.e. H_{L1}=H_{G1} and W_{L1}=W_{G1}) as well as an aperture of camera **200**, such that the optical height and the optical width of lens element L₁ represent also the aperture height (HA) and aperture width (WA) of lens **204** respectively. The D-cut of L1 and G1 means that also STC **200**'s aperture changes accordingly, such that the aperture is not axial symmetric. The cutting allows for a small lens heights H_{G1}, which are required for small MH_Ms, and still relatively large effective aperture diameters (DAs) which satisfy DA>H_{G1}.

In other examples, an EFL of lens **204** may be 8 mm-50 mm.

G2 is D-cut as well. The optical height (H_{L3}) and width (W_{L3}) of lens element L₃ may define the optical height, width and aperture of G2. Prism **202** is D-cut as well.

Detailed optical data and surface data are given in Tables 2-3 for the example of the lens elements in FIG. 8. The values provided for these examples are purely illustrative and according to other examples, other values can be used.

Surface types are defined in Table 4. The coefficients for the surfaces are defined in Table 5. The surface types are:

- a) Plano: flat surfaces, no curvature
- b) Q type 1 (QT1) surface sag formula:

$$z(r) = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + D_{con}(u) \quad (\text{Eq. 1})$$

$$D_{con}(u) = u^4 \sum_{n=0}^N A_n Q_n^{con}(u^2)$$

$$u = \frac{r}{r_{norm}}, x = u^2$$

$$Q_0^{con}(x)=1 \quad Q_1^{con}=-(5-6x) \quad Q_2^{con}=15-14x(3-2x)$$

$$Q_3^{con}=-\{35-12x[14-x(21-10x)]\}$$

$$Q_4^{con}=70-3x\{168-5x[84-11x(8-3x)]\}$$

$$Q_5^{con}=-[126-x\{1260-11x\{420-x[720-13x(45-14x)]\}}]$$

- c) Even Asphere (ASP) surface sag formula:

$$z(r) = \frac{cr^2}{1 + \sqrt{1 - (1+k)c^2r^2}} + \alpha_1 r^2 + \alpha_2 r^4 + \alpha_3 r^6 + \alpha_4 r^8 + \alpha_5 r^{10} + \alpha_6 r^{12} + \alpha_7 r^{14} + \alpha_8 r^{16} \quad (\text{Eq. 2})$$

Example 800

EFL = 17.37 mm, f number = 2.35 (Eff. DA/2 = 3.7 mm), HFOV = 12.8 deg.

Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S.	Plano	Infinity	-1.543	4.000				
2	Lens 1	ASP	5.876	3.001	4.000	Glass	1.48	84.1	10.424

-continued

Example 800									
EFL = 17.37 mm, f number = 2.35 (Eff. DA/2 = 3.7 mm), HFOV = 12.8 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
3			-30.470	0.373	3.666				
4	Lens 2	ASP	-34.332	2.433	3.515	Plastic	1.61	25.6	-14.027
5			11.925	0.892	2.892				
6	Lens 3	ASP	-30.326	2.043	2.774	Plastic	1.67	19.2	14.887
7			-7.770	0.035	2.782				
8	Lens 4	ASP	-22.896	1.591	2.672	Plastic	1.54	55.9	24.980
9			-8.758	0.224	2.781				
10	Lens 5	ASP	-69.824	1.017	2.618	Plastic	1.61	25.6	-110.272
11			2962.171	0.927	2.516				
12	Lens 6	ASP	-4.642	0.347	2.497	Plastic	1.61	25.6	-8.581
13			-38.037	5.279	2.732				
14	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.2	
15			Infinity	0.350	—				
16	Image	Plano	Infinity	—	—				

where $\{z, r\}$ are the standard cylindrical polar coordinates, c is the paraxial curvature of the surface, k is the conic parameter, r_{norm} is generally one half of the surface's clear aperture, and A_n are the polynomial coefficients shown in lens data tables. The Z axis is positive towards the image. Values for aperture radius are given as a clear aperture radius, i.e. DA/2. The reference wavelength is 555.0 nm. Units are in mm except for refraction index ("Index") and Abbe #.

The same formulas, units and definitions are used also for Tables 6-11.

TABLE 5

Aspheric Coefficients					
Surface #	Conic	4 th	6 th	8 th	10 th
2	0	-2.45E-06	1.03E-05	-9.26E-07	8.22E-08
3	0	-7.38E-05	1.77E-06	-1.09E-06	1.02E-07
4	0	-1.19E-03	-1.87E-05	1.97E-06	1.94E-07
5	0	-2.78E-04	-2.26E-04	-8.07E-06	2.41E-06
6	0	1.31E-03	-5.27E-04	7.78E-06	-2.89E-06
7	0	5.97E-04	-2.77E-04	-1.68E-05	5.15E-06
8	0	-2.20E-03	1.74E-04	-3.81E-05	1.19E-06
9	0	-1.95E-03	-1.73E-04	-6.31E-06	3.10E-07
10	0	1.16E-04	-8.54E-04	3.38E-05	8.06E-06
11	0	4.28E-04	-7.22E-04	4.63E-05	3.53E-06
12	0	-3.11E-03	-1.03E-04	-6.08E-06	3.71E-07
13	0	-3.09E-03	1.09E-04	8.66E-06	-2.01E-06
Aspheric Coefficients (Continued)					
Surface #	12 th	14 th	16 th		
2	-2.68E-09	5.28E-13	-2.97E-14		
3	-1.97E-09	-7.40E-14	9.91E-15		
4	-7.91E-09	3.55E-12	4.10E-13		
5	-8.91E-08	1.09E-09	8.66E-13		
6	5.28E-07	-1.95E-08	-4.33E-13		
7	-2.86E-07	7.01E-09	6.35E-12		
8	1.26E-07	-6.82E-09	-5.07E-12		
9	2.95E-07	-3.40E-08	1.09E-09		
10	-1.52E-07	-2.21E-08	1.29E-10		
11	2.90E-08	-2.34E-08	6.33E-10		
12	2.55E-08	-6.72E-08	5.93E-09		
13	-1.59E-07	3.54E-08	-1.12E-09		

FIG. 9 shows optical lens system 900 included in STC 250 from FIGS. 2C-D in a cross-sectional view and with ray-tracing. ΔL_O is 2.7 mm, ΔC is 0.15 mm.

The optical height (H_{L1}) and width (W_{L1}) of lens element L_1 may define the optical height and width of lens 254 as well as an aperture of STC 250, such that the optical height and the optical width of lens element L_1 represent also the aperture height (HA) and aperture width (WA) of lens 254 respectively. The D-cut of L_1 means that also STC 250's aperture changes accordingly. The cutting allows for a small HA and still relatively large effective DAs which satisfy DA>HA. In other examples, an EFL of lens 254 may be 8 mm-50 mm. Prism 252 is D-cut as well. A s-FOV_T is 69.5deg×42.58deg, i.e. a horizontal direction of s-FOV_T ("H-s-FOV_T") is H-s-FOV_T=69.5deg, a vertical direction of s-FOV_T ("V-s-FOV_T") is V-s-FOV_T=42.58deg. s-FOV_T covers a 16:9 FOV ratio of a FOV_H=82deg (diagonal) of a Wide camera that may be included in a mobile device together with the STC.

Detailed optical data and surface data are given in Tables 6-7.

TABLES 6-7

Aspheric Coefficients					
Surface #	Conic	4 th	6 th	8 th	10 th
2	0	-3.85E-04	-1.13E-05	-2.76E-06	-9.98E-08
3	0	8.43E-04	-5.11E-05	-4.46E-07	2.27E-07
4	0	-1.17E-03	1.34E-06	-2.87E-06	1.51E-06
5	0	2.58E-03	1.86E-05	6.27E-05	8.43E-06
6	0	2.92E-03	-3.50E-04	5.10E-05	7.31E-06
7	0	9.39E-04	-1.77E-04	-2.83E-05	3.52E-06
8	0	1.43E-03	-6.10E-04	-1.44E-04	-1.32E-05
9	0	-3.91E-03	-4.05E-04	1.55E-05	-1.41E-06
10	0	-2.93E-03	-5.10E-04	2.28E-04	3.56E-05
11	0	5.67E-03	-4.18E-04	9.28E-05	1.35E-05
12	0	-3.50E-03	-7.08E-04	5.45E-04	-2.00E-04
13	0	-1.21E-02	9.04E-04	-9.24E-05	-7.71E-06
Aspheric Coefficients (Continued)					
Surface #	12 th	14 th	16 th		
2	-5.82E-10	7.38E-10	-1.79E-10		
3	-8.02E-09	-1.13E-09	6.18E-11		
4	1.92E-07	-6.49E-09	-1.47E-09		
5	2.83E-07	1.11E-07	-2.77E-08		
6	1.28E-06	-1.04E-07	-1.23E-08		
7	8.01E-07	5.64E-07	-6.36E-08		
8	-2.77E-06	-5.16E-07	9.10E-08		
9	-1.75E-06	-4.88E-07	5.63E-08		
10	9.70E-07	-9.49E-07	4.30E-08		

TABLES 6-7-continued

11	-9.57E-07	9.48E-07	-1.14E-07
12	-1.59E-05	7.95E-06	-4.71E-07
13	-5.10E-06	9.72E-07	-3.87E-08

height and the optical width of lens element L_1 represent also the aperture height (HA) and aperture width (WA) of lens **1004** respectively. Lens **1004**, i.e. L_1 and further lens elements, as well as prism **1002** are D-cut. In other examples, an EFL of lens **1004** may be 8 mm-50 mm and SD may be

Example 900									
EFL = 14.1 mm, f number = 2.45 (Eff. DA/2 = 2.9 mm), HFOV = 11.5 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S.	Plano	Infinity	-1.246	3.000				
2	Lens 1	ASP	3.981	1.941	3.013	Glass	1.48	84.1	7.619
3			-44.268	1.288	2.903				
4	Lens 2	ASP	23.035	0.281	2.448	Plastic	1.61	25.6	-8.410
5			4.224	0.020	2.232				
6	Lens 3	ASP	3.832	0.531	2.251	Plastic	1.53	55.7	32.268
7			4.684	1.504	2.199				
8	Lens 4	ASP	43.147	0.869	2.230	Plastic	1.66	20.4	7.554
9			-5.655	0.018	2.326				
10	Lens 5	ASP	-9.456	0.238	2.231	Plastic	1.61	25.6	-8.644
11			12.423	3.160	2.100				
12	Lens 6	ASP	-3.038	1.058	2.193	Plastic	1.53	55.7	-14.591
13			-5.570	0.941	2.654				
14	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.2	
15			Infinity	0.350	—				
16	Image	Plano	Infinity	—	—				

FIG. 10 shows another optical lens system **1000** disclosed herein in a cross-sectional view and with ray-tracing. Optical

TABLE 8

Example 1000									
EFL = 14.1 mm, f number = 2.45 (Eff. DA/2 = 2.9 mm), HFOV = 15.7 deg.									
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length
1	A.S.	Plano	Infinity	-1.121	3.000				
2	Lens 1	ASP	4.321	2.708	3.018	Glass	1.48	84.1	9.046
3			208.561	0.539	2.844				
4	Lens 2	ASP	23.819	0.709	2.712	Plastic	1.57	37.4	13.890
5			-11.728	0.062	2.688				
6	Lens 3	ASP	-8.405	0.308	2.646	Plastic	1.61	25.6	-6.440
7			7.678	0.811	2.474				
8	Lens 4	ASP	-58.073	1.934	2.505	Plastic	1.67	19.2	8.073
9			-5.062	0.038	2.826				
10	Lens 5	ASP	-35.454	1.794	2.765	Plastic	1.61	25.6	-11.971
11			9.540	1.157	2.577				
12	Lens 6	ASP	-67.401	0.525	2.551	Plastic	1.67	19.2	-15.109
13			12.096	3.371	2.909				
14	Filter	Plano	Infinity	0.210	—	Glass	1.52	64.2	
15			Infinity	0.350	—				
16	Image	Plano	Infinity	—	—				

lens system **1000** may be included in a STC such as STC **200** or STC **250**. Optical lens system **1000** includes a prism **1002**, a lens **1004** including N=6 lens elements, an (optional) optical filter **1005** and an image sensor **1006**. ΔL_O is 2.7 mm, ΔC is 0.15 mm. A distance from a first rotation axis to OPFE **1002**'s light exiting surface is 0.5 mm. A distance from a second rotation axis to OPFE **1000**'s light exiting surface is 4.3 mm.

H_{L1} and W_{L1} of lens element L_1 may define the optical height and width of lens **1004** as well as an aperture of a STC that includes optical lens system **1000**, such that the optical

TABLE 9

Surface #	Aspheric Coefficients				
	Conic	4 th	6 th	8 th	10 th
2	0	-1.30E-04	-1.20E-05	-1.85E-06	1.87E-07
3	0	7.73E-04	-2.67E-04	-5.52E-06	2.38E-06
4	0	-7.46E-04	-4.36E-04	-1.63E-05	5.83E-06
5	0	-1.90E-03	2.66E-05	-1.30E-06	1.60E-06
6	0	1.67E-03	-5.80E-04	8.83E-05	-5.04E-06
7	0	3.06E-03	-2.99E-04	-5.92E-05	3.02E-05
8	0	-1.90E-03	3.91E-04	-1.26E-04	5.48E-06
9	0	-1.56E-03	2.81E-04	-1.86E-05	-6.04E-06

TABLE 9-continued

10	0	-3.81E-03	-1.27E-04	1.13E-04	-1.19E-05
11	0	-3.87E-03	-7.18E-04	1.83E-04	-3.44E-05
12	0	-2.57E-02	1.22E-03	-4.89E-05	-4.84E-06
13	0	-2.37E-02	2.34E-03	-1.49E-04	-1.79E-06

Aspheric Coefficients (Continued)					
Surface #		12 th	14 th	16 th	
2		-5.47E-08	3.88E-09	-2.08E-10	
3		-1.56E-07	8.60E-09	-2.61E-10	
4		2.11E-08	-4.48E-08	1.98E-09	
5		-9.67E-07	9.65E-08	-2.76E-09	
6		5.46E-07	-1.04E-07	4.93E-09	
7		-3.15E-06	3.96E-07	-2.26E-08	
8		1.13E-06	-1.54E-07	5.51E-09	
9		1.08E-06	-7.98E-08	2.33E-09	
10		2.81E-07	1.09E-07	-6.13E-09	
11		4.58E-06	-4.02E-07	1.43E-08	
12		-2.96E-06	6.02E-07	-2.54E-08	
13		1.11E-06	-5.00E-08	4.36E-10	

FIG. 11 shows another optical lens system **1100** disclosed herein with ray-tracing. FIG. 11A shows optical lens system **1100** in a cross-sectional view. FIG. 11B shows optical lens system **1100** in a top view. FIG. 11C shows optical lens system **1100** in a perspective view.

Optical lens system **1100** includes an OPFE **1102** (e.g. a prism or a mirror), a lens **1104** including N=6 lens elements L₁-L₆, an (optional) optical filter **1105** and an image sensor **1106**. Lens **1104** has an optical axis **1108**. Lens **1104** is a cut lens. The cutting is performed such that a height of lens **1104** ("H_L", measured along the y-axis) is 5.1 mm, as shown in

FIG. 11A. This means that a cut ratio, i.e. a ratio between which a height of a lens element differs from its width, is 20% or less. Cutting a lens like lens **1104** is beneficial as of two desired outcomes: it reduces a height of the cut lens itself (what reduces MH_S) and it reduces a height of an OPFE such as OPFE **1102** (what reduces MH_M). Specifically, cutting a lens by X % will reduce MH_M and MH_S by about 0.5·X %-X %. For example, cutting a lens by 20% will reduce MH_M and MH_S by about 10%-20%.

In FIG. 11C is visible that image sensor **1106** is oriented in an anti-parallel STC sensor configuration. Orienting image sensor **1106** in an anti-parallel STC sensor configuration is beneficial as it aligns a fast scan axis of OPFE **1102** with a longer side of s-FOV_T and a slow scan axis of OPFE **1102** with a shorter side of s-FOV_T. Therefore, a maximum rotational movement required for covering s-FOV_T can be smaller than in STCs that use a parallel STC sensor configuration. For example and for covering a 16:9 ratio of FOV_w=82° (diagonal), for STC **1250** (anti-parallel STC sensor configuration) a maximum rotational movement of OPFE **1102** is 21.35° (see Table 12), whereas for STC **250** and optical lens system **1000** (parallel STC sensor configuration) a maximum rotational movement of OPFE **252** and OPFE **1002** is 29.8° (see Table 2) and 25.5° (see Table 3) respectively. Smaller maximum rotational movements are beneficial, as they can be provided by simpler actuators as well as simpler and more accurate actuation control.

Detailed optical data and surface data are given in Tables 10-11. An effective f/#based on an effective lens aperture diameter as known in the art is given.

TABLE 10

Example 1100										
EFL = 14.1 mm, Eff. f number = 2.43 (Eff. DA/2 = 2.9 mm), HFOV = 13.7 deg.										
Surface #	Comment	Type	Curvature Radius	Thickness	Aperture Radius (D/2)	Material	Index	Abbe #	Focal Length	
1	A.S.	Plano	Infinity	-1.058	3.000					
2	Lens 1	ASP	4.430	1.600	3.001	Glass	1.48	84.1	9.556	
3			85.664	1.088	2.967					
4	Lens 2	ASP	8.627	0.819	2.773	Plastic	1.53	55.69	12.034	
5			-24.809	0.109	2.716					
6	Lens 3	ASP	-8.650	0.503	2.691	Plastic	1.61	25.59	-4.544	
7			4.256	0.517	2.414					
8	Lens 4	ASP	7.547	1.600	2.412	Plastic	1.67	19.24	8.497	
9			-22.121	0.037	2.520					
10	Lens 5	ASP	5.421	0.788	2.424	Plastic	1.61	25.59	-15.577	
11			3.276	2.994	2.293					
12	Lens 6	ASP	-17.791	0.683	2.797	Plastic	1.53	55.69	-81.994	
13			-30.272	2.382	3.001					
14	IR Filter	Plano	Infinity	0.210	—	Glass	1.52	64.17		
15			Infinity	0.350	—					
16	Image	Plano	Infinity	—	—					

TABLE 11

Aspheric Coefficients									
Surface #	Conic	4 th	6 th	8 th	10 th	12 th	14 th	16 th	
2	0	-5.19E-04	-1.78E-05	-4.48E-06	-8.07E-08	-1.64E-08	-8.77E-10	1.34E-10	
3	0	-1.12E-03	2.33E-04	-4.20E-05	4.57E-06	-4.19E-07	2.34E-08	-6.24E-10	
4	0	-6.44E-03	6.51E-04	-4.40E-05	2.04E-06	8.43E-08	7.98E-09	-1.44E-09	
5	0	-2.21E-04	4.15E-04	3.01E-05	1.90E-05	-3.57E-06	1.45E-07	4.15E-09	
6	0	7.41E-03	-1.55E-03	2.67E-04	-2.54E-05	2.21E-06	-2.67E-07	1.70E-08	
7	0	-5.36E-03	1.97E-04	-3.28E-05	-1.36E-06	1.91E-06	-4.06E-07	4.19E-08	
8	0	1.57E-03	-1.49E-03	2.62E-04	-8.49E-05	1.59E-05	-2.24E-06	1.37E-07	
9	0	-8.09E-03	5.64E-04	-3.26E-04	7.59E-05	-9.18E-06	5.63E-07	-1.18E-08	

TABLE 11-continued

Surface #	Conic	Aspheric Coefficients						
		4 th	6 th	8 th	10 th	12 th	14 th	16 th
10	0	-3.63E-02	2.08E-03	2.51E-04	-2.96E-05	2.33E-06	-2.20E-07	8.14E-09
11	0	-2.75E-02	1.60E-03	9.49E-04	-3.33E-04	5.92E-05	-5.44E-06	2.06E-07
12	0	-1.02E-02	2.26E-04	-1.51E-04	5.00E-05	-8.69E-06	8.68E-07	-3.28E-08
13	0	-9.62E-03	-1.91E-05	1.57E-05	-3.19E-06	4.16E-07	-3.72E-08	2.12E-09

With reference to FIG. 7, a s-FOV_T of STC 1250 covers 69.4°×42.6°. It is noted that s-FOV_T covers a 16:9 ratio of FOV_W of a Wide (or Main) camera such as 130 having a (diagonal) FOV_W=82°. Table 12 provides the rotation values of OPFE 1102 around a first and a second rotation respectively that are required for scanning to the 9 respective n-FOV_Ts shown in FIG. 7. The values refer to a scanning action that starts from n-FOV_T 5, i.e. the (Center, Center) position. For example for scanning n-FOV_T to n-FOV_T 9 or (Bottom, Right), starting from (Center, Center) position n-FOV_T 5, OPFE 1102 must be rotated by -13.18 degrees around the first rotation axis and by -8.9 degrees around the second rotation axis.

TABLE 12

	Left	Center	Right
Top	(9.2, 21.35)	(12.05, 0)	(9.2, -21.35)
Center	(-1.99, 17)	(0, 0)	(-1.99, -17)
Bottom	(-13.18, 8.9)	(-12.05, 0)	(-13.18, -8.9)

FIG. 12 shows a mobile device 1200 (e.g. a smartphone) including a STC 1250 in a cross-sectional view. STC 1250 may include optical lens system 1100 shown in FIGS. 11A-C. Mobile device 1200 has a front surface 1202 (e.g. including a screen, not shown) and a rear surface 1210 including STC 1250's aperture 1259. Mobile device 1200 has a regular region 1212 of thickness "T" and a camera bump region 1214 that is elevated by a height B over regular region 1212. Bump region 1214 has a bump length ("L_B") and a bump thickness T+B. For achieving short L_B, a module region 1252 of STC 1250 having a height MH_M is included in bump region 1214 and shoulder region 1254 of STC 1250 having a height MH_S<MH_M is included in regular region 1312. This means that OPFE 1102 is included in bump region 1214 and lens 1104 and image sensor 1106 are included in regular region 1212. Optionally, in some embodiments, parts of shoulder region 1254 may also be included in bump region 1214.

FIGS. 13A-B show OPFE 1102 (here, a prism) of STC 1250 in a zero scan position. FIG. 13A shows OPFE 1102 in a side (or cross-sectional) view. FIG. 13B shows OPFE 1102 in a perspective view.

OPFE 1102 has a light entering surface 1302 and a light exiting surface 1304. The location of first rotation axis 1306 and second rotation axis 1312 are shown. OPFE 1102 has a prism height ("H_P") and an optical (or optically active) prism height ("H_{P-O}"), a prism length ("L_P") and an optical prism length ("L_{P-O}") and a prism width ("W_P"), as shown.

A distance from first rotation axis 1306 to OPFE 1102's light exiting surface 1304 is Δ1306. Here, Δ1306=0.5 mm and a ratio of Δ1306 and the prism length L_P is Δ1306/L_P=0.07. This de-center location of OPFE 1102 is beneficial for minimizing MH_M. A distance from second rotation axis

1312 to OPFE 1102's light entering surface 1302 is Δ1312. Here, Δ1312=3.35 mm and a ratio of Δ1312 and the prism height is Δ1312/H_P=0.55.

OPFE 1102 has a non-cut center axis 1332 that indicates a center of a non-cut OPFE 1102 with respect to the y-axis. OPFE 1102 has a cut center axis 1334 that indicates a center of cut OPFE 1102 with respect to the y-axis. Both first rotation axis 1306 and second rotation axis 1312 intersect with optical axis 1108 of lens 1104 and with non-cut center axis 1332. In other words and referring to FIG. 4D and FIGS. 6A-D, in optical lens system 1100 ΔC=0.

OPFE 1102 includes an exiting-surface top stray light prevention mask 1322 having a height H_{T-SM}, an exiting-surface bottom stray light prevention mask 1324 having a height H_{B-SM}, an entering-surface left stray light prevention mask 1326 having a length LL-SM and an entering-surface right stray light prevention mask 1328 having a length L_{R-SM}. Values and ranges are given in Table 13 in mm. The stray light prevention masks are beneficial because they prevent stray light from reaching an image sensor such as image sensor 1106. Stray light is undesired light emitted or reflected from an object in a scene which enters a camera's aperture and reaches an image sensor at a light path that is different from a planned (or desired) light path. A planned light path is described as follows:

1. Light is emitted or reflected by an object in a scene.
2. Light enters a camera's aperture.
3. For examples where the OPFE is a mirror, light is reflected once at the mirror's surface. For examples where the OPFE is a prism, light passes once a light entering surface of the prism, is reflected once at the prism's reflective surface, and then passes once a light exiting surface of the prism.
4. Light passes once all surfaces of a lens.
5. Light impinges on an image sensor.

Light that reaches an image sensor on any light path other than the planned light path described above is considered undesired and referred to as stray light.

Values and ranges are given in Table 13 in mm.

L_{P-O}/L_P=0.76, i.e. left stray light prevention mask 1326 and right stray light prevention mask 1328, which are located at the light entering surface 1302, together cover a surface area of more than 20% and less than 30% of the area of the light entering surface 1302. H_{P-O}/H_P=0.83, i.e. top stray light prevention mask 1322 and bottom stray light prevention mask 1324 which are located at the light exiting surface 1304, together cover a surface area of more than 10% and less than 20% of the area of the light entering surface 1304.

TABLE 13

	Value	Value range
H _P	6.11	3-10
H _{P-O}	5.1	2-10

TABLE 13-continued

	Value	Value range
L_P	6.72	3-12
$L_{P,O}$	5.13	2-12
W_P	10.3	4-15
H_{T-SM}	0.81	0.1-2.5
H_{B-SM}	0.2	0.05-2.5
L_{L-SM}	0.83	0.1-4
L_{R-SM}	0.76	0.1-4
$\Delta 1306$	0.5	0.2-3
$\Delta 1312$	3.35	1.5-6

Table 14 summarizes values and ratios thereof of various features that are included in STC **200**, STC **250** and STC **1230** and optical lens systems **800**, **900**, **1000** and **1100**. H_{G1} , W_{G1} , H_{G2} , W_{G2} , ΔC , HA , WA , DA , HA_{G2} , WA_{G2} , DA_{G2} , H_P , W_P , L_P , ΔLO , TTL , BFL , EFL , EFL_{G1} , EFL_{G2} , SD , H_{sensor} , MH_S , MH_M , H_S , H_M , ALT , ALT_{G1} , ALT_{G2} , T_1 , f_1 are given in mm. n-FOV_T, S-FOV_T, α -OPFE and β -OPFE are given in degrees.

In other examples, the values may differ from the values given here by e.g. $\pm 10\%$, or by $\pm 20\%$, or by even $\pm 30\%$.

“Type” specifies whether the optical lens system uses a parallel STC sensor configuration (“P”) or an anti-parallel STC sensor configuration (“A-P”).

“16:9 W ratio” indicates whether the s-FOV_T of the respective optical lens system covers (“Y”) or not covers (“N”) a 16:9 ratio of a Wide camera having a diagonal FOV_W=82°.

DA is the aperture diameter. For cut lenses, an effective aperture diameter is given. “Effective aperture diameter” means here a diameter of a circular (or axial symmetric) aperture, wherein the circular aperture has a same aperture area as the cut lens (which has a non axial-symmetric aperture).

EFL_{G1} and EFL_{G2} are the effective focal lengths of lens groups G1 and G2 respectively.

The average lens thickness (“ALT”) measures the average thickness of all lens elements. ALT_{G1} and ALT_{G2} is the ALT of G1 and G2 respectively.

T_1 is the center thicknesses of L_1 . f_1 is the focal length of L_1 .

All other parameters not specifically defined here have their ordinary meaning as known in the art.

TABLE 14

Parameter	800	900	1000	1100	Explanation
Type	P	P	P	A-P	Sensor configuration
H_{G1}	6.00	5.00	5.00	5.10	Optical height of G1
W_{G1}	8.00	6.00	6.00	6.00	Optical width of G1
H_{G2}	4.40	—	—	—	Optical height of G2
W_{G2}	5.56	—	—	—	Optical width of G2
ΔC	0.15	0.15	0.15	0.00	Lens-OPFE de-center
HA	6.00	5.00	5.00	5.10	Aperture height of lens
WA	8.00	6.00	6.00	6.00	Aperture width of lens
DA	7.40	5.75	5.75	5.80	Aperture diameter of lens
HA_{G2}	4.40	—	—	—	HA of G2
WA_{G2}	5.56	—	—	—	WA of G2
DA_{G2}	5.24	—	—	—	DA of G2
H_P	6.80	6.80	6.80	6.11	Height of prism
W_P	13.50	11.00	11.00	10.30	Width of prism
L_P	7.70	7.70	7.70	6.72	Length of prism
ΔLO	2.70	2.70	2.70	3.10	Distance lens-OPFE
TTL	18.72	12.41	14.52	13.68	
BFL	5.84	1.50	3.93	2.94	
EFL	17.37	14.10	14.10	14.10	
EFL_{G1}	21.47	—	—	—	EFL of G1
EFL_{G2}	1969.60	—	—	—	EFL of G2
f number	2.35	2.45	2.45	2.43	
n-FOV _T	25.6°	22.8°	31°	27.40°	Diagonal n-FOV _T
s-FOV _T	50.9° x	69.5° x	69.5° x	69.4° x	
	32.5°	42.58°	42.58°	42.6°	
16:9 W ratio	N	Y	Y	Y	
α -OPFE	$\pm 7.85^\circ$	± 11.67	± 9.67	± 13.18	Maximal rotation around 1 st rotation axis (402, 452, 1306)
β -OPFE	$\pm 21.67^\circ$	± 29.80	± 25.54	± 21.35	Maximal rotation around 2 nd rotation axis (502, 552, 1312)
SD	8.00	5.60	8.00	7.00	Image sensor diagonal
H_{sensor}	4.80	1.68	2.40	5.60	Sensor height
MH_M	8.24	8.84	8.48	7.48	Minimum module height
MH_S	6.16	5.95	6.18	5.34	Minimum shoulder height
H_M	9.74	10.34	9.98	8.98	Module height
H_S	7.66	7.45	7.68	6.84	Shoulder height
ALT	1.74	0.82	1.33	1.00	Average thickness of lens elements L1-L6
ALT_{G1}	2.72	—	—	—	ALT of G1
ALT_{G2}	1.25	—	—	—	ALT of G2
T_1	3.00	1.94	2.71	1.60	Center thickness of L_1
f_1	10.42	7.62	9.05	9.56	Focal length of L_1
$\Delta C/HA$	0.025	0.030	0.030	0.000	
$\Delta C/H_S$	0.02	0.020	0.020	0.000	
$\Delta C/H_O$	0.022	0.022	0.022	0.000	
D-cut ratio (Lens, G1)	0.75	0.83	0.83	0.85	=HA/WA

TABLE 14-continued

Parameter	800	900	1000	1100	Explanation
D-cut ratio (G2)	0.79	—	—	—	$=H_{G2}/W_{G2}$
D-cut ratio (OPFE)	0.88	0.88	0.88	0.91	$=H_0/L_0$
EFL/TTL	0.93	1.14	0.97	1.03	
BFL/EFL	0.34	0.11	0.28	0.21	
BFL/TTL	0.31	0.12	0.27	0.21	
DA/DA _{G2}	1.41	—	—	—	
DA/H _S	0.97	0.77	0.75	0.85	
WA/H _S	1.04	0.81	0.78	0.88	
DA/H _M	0.76	0.56	0.58	0.65	
H _{G1} /H _{G2}	1.37	—	—	—	
H _{G1} /MH _S	0.97	0.84	0.81	0.96	
H _{G1} /MH _M	0.73	0.57	0.59	0.68	
H _{G1} /H _S	0.78	0.67	0.65	0.75	
H _{G1} /H _M	0.62	0.48	0.50	0.57	
H _S /H _M	0.79	0.72	0.77	0.76	
SD/EFL	0.46	0.40	0.57	0.50	
T ₁ /ALT	1.73	2.37	2.04	1.60	
ALT _{G1} /ALT	1.56	—	—	—	
ALT _{G2} /ALT	0.72	—	—	—	
ALT _{G1} /ALT _{G2}	2.17	—	—	—	
f ₁ /EFL	0.60	0.54	0.64	0.68	
ΔLO/TTL	0.14	0.22	0.19	0.23	

While this disclosure has been described in terms of certain embodiments and generally associated methods, alterations and permutations of the embodiments and methods will be apparent to those skilled in the art. The disclosure is to be understood as not limited by the specific embodiments described herein, but only by the scope of the appended claims.

Furthermore, for the sake of clarity the term “substantially” is used herein to imply the possibility of variations in values within an acceptable range. According to one example, the term “substantially” used herein should be interpreted to imply possible variation of up to 5% over or under any specified value. According to another example, the term “substantially” used herein should be interpreted to imply possible variation of up to 2.5% over or under any specified value. According to a further example, the term “substantially” used herein should be interpreted to imply possible variation of up to 1% over or under any specified value.

All references mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual reference was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present disclosure.

What is claimed is:

1. A scanning Tele camera (STC), comprising:

a single optical path folding element (OPFE) for folding a first optical path OP1 that defines a height H axis to a second optical path OP2 that defines a length L axis; an OPFE actuator;

a lens having a lens optical axis parallel to OP2, an effective focal length EFL in the range of 8 mm-50 mm, and a f number $f/\# < 3.5$; and

an image sensor,

wherein the STC has a STC native field-of-view n-FOV_T, wherein the OPFE actuator is configured to rotate the OPFE around a first rotation axis and around a second rotation axis for scanning a scene with the n-FOV that provides an effective Tele scanning FOV s-FOV_T,

wherein the s-FOV_T has a longer side and a shorter side, and

wherein the longer side of s-FOV_T is greater than 40 degrees.

2. The STC of claim 1, wherein the first rotation axis is perpendicular to both OP1 and OP2, and wherein the second rotation axis is parallel to OP1.

3. The STC of claim 1, wherein the OPFE has an OPFE length L_O measured along an axis parallel to OP2, an OPFE light entering surface and an OPFE light exiting surface, wherein the first rotation axis is located at a distance Δ1 from the light exiting surface of the OPFE, and wherein $\Delta1/L_O < 0.25$.

4. The STC of claim 3, wherein $\Delta1/L_O < 0.15$.

5. The STC of claim 1, wherein $f/\# < 3$.

6. The STC of claim 1, wherein $f/\# < 2.5$.

7. The STC of claim 1, wherein the STC is included in a dual-camera that comprises a Wide camera having a Wide camera field of view FOV_W, and wherein a center location of s-FOV_T is identical with a center location of FOV_W.

8. The STC of claim 7, wherein FOV_W is in the range of 50-120 degrees, and wherein s-FOV_T covers a 16:9 segment of FOV_W.

9. The STC of claim 7, wherein FOV_W is in the range of 75-85 degrees, and wherein s-FOV_T covers a 16:9 segment of FOV_W.

10. The STC of claim 1, wherein the longer side of s-FOV_T is greater than 45 degrees.

11. The STC of claim 1, wherein the longer side of s-FOV_T is greater than 50 degrees.

12. The STC of claim 1, wherein the shorter side of s-FOV_T is greater than 20 degrees.

13. The STC of claim 1, wherein the shorter side of s-FOV_T is greater than 25 degrees.

14. The STC of claim 1, wherein the shorter side of s-FOV_T is greater than 30 degrees.

15. The STC of claim 1, wherein the rotation of the OPFE along the first rotation axis is by more than ±5 degrees around a zero scan position.

16. The STC of claim 1, wherein the rotation of the OPFE along the second rotation axis is by more than ±15 degrees around a zero scan position.

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17. The STC of claim 1, wherein the OPFE is a prism.

18. The STC of claim 1, wherein the lens has a maximum lens aperture height H_A measured along OP1, wherein the STC is included in a camera module, wherein the camera module is divided into a module region having a module region height H_M and a shoulder region having a shoulder region height $H_S < H_M$, all heights being measured along OP1, and wherein $H_S < H_A + 3$ mm.

19. The STC of claim 1, wherein the STC is included in a mobile device.

20. The STC of claim 19, wherein the mobile device is a smartphone.

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